

Tuesday recap

Kahoot-style quiz

**12 quick questions on Tuesday's
Mendel vocab and crosses
Not graded
Fast and correct wins.**



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-tg1-quiz](https://greymonroe.github.io/live-poll-test/quiz-play.html?quiz=pls152-tg1-quiz)

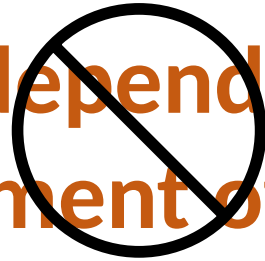
[host screen](#)

Transmission Genetics 2: Breaking Mendel's Rules

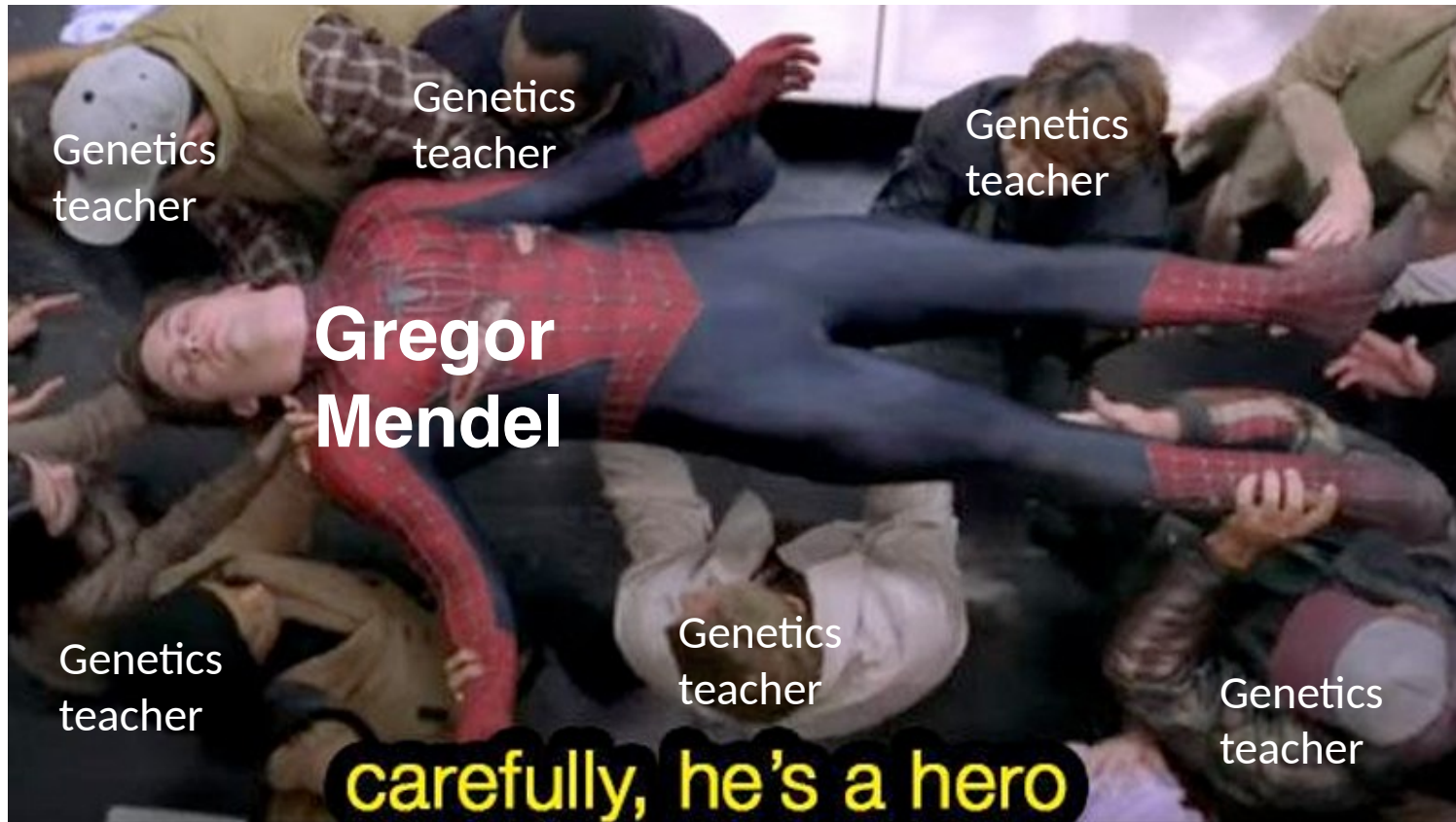
Dominance



Independent
assortment of genes



Random
segregation of
alleles





1. **Epistasis:** A form of gene interaction where one gene masks or modifies the effect of another gene at a different locus.
2. **Recessive epistasis:** When the recessive allele of one gene masks the effects of either allele of another gene.
3. **Dominant Epistasis:** When the dominant allele of one gene masks the effects of either allele of another gene.
4. **Synthetic lethality:** A situation where the combination of mutations in two or more genes leads to death, whereas a mutation in any single gene does not.
5. **Heterosis:** Also known as hybrid vigor, refers to the phenomenon where hybrid offspring exhibit superior phenotypic characteristics compared to their parents.
6. **Pleiotropy:** The phenomenon where a single gene affects multiple, seemingly unrelated phenotypic traits.
7. **Linkage (disequilibrium):** The tendency of genes located close together on the same chromosome to be inherited together.
8. **Recombination:** The process by which genetic material is physically mixed during sexual reproduction, leading to the production of offspring with combinations of traits that differ from either parent.
9. **Chi-squared test:** A statistical test used to determine if observed data fits expected frequencies, often used in genetics to test for independent assortment or linkage.
10. **Marker:** A known DNA sequence or gene on a chromosome that can be used to identify a particular location or linked genes.
11. **Incomplete dominance:** A genetic situation in which one allele does not fully dominate another allele, resulting in an intermediate phenotype.
12. **Additive genetic variation:** The portion of phenotypic variation in a population that is due to the additive effects of alleles (i.e., the sum of individual effects).
13. **Co-dominance:** A genetic scenario where both alleles for a gene are expressed in a heterozygous individual, leading to a phenotype that shows both traits simultaneously.
14. **Overdominance:** A situation where the heterozygote has a phenotype that is more extreme or better adapted than either homozygote.
15. **Heterozygous advantage:** When heterozygous individuals have a higher fitness than either type of homozygous individual.
16. **Lethal alleles:** Alleles that cause the death of the organism that carries them, often before the organism can reproduce.
17. **Meiotic drive:** A phenomenon where certain alleles are transmitted to offspring more frequently than would be expected by Mendelian inheritance.
18. **Gene drive:** A genetic engineering technology used to spread a particular genetic modification throughout a population rapidly, by ensuring that a particular allele is preferentially transmitted to offspring.

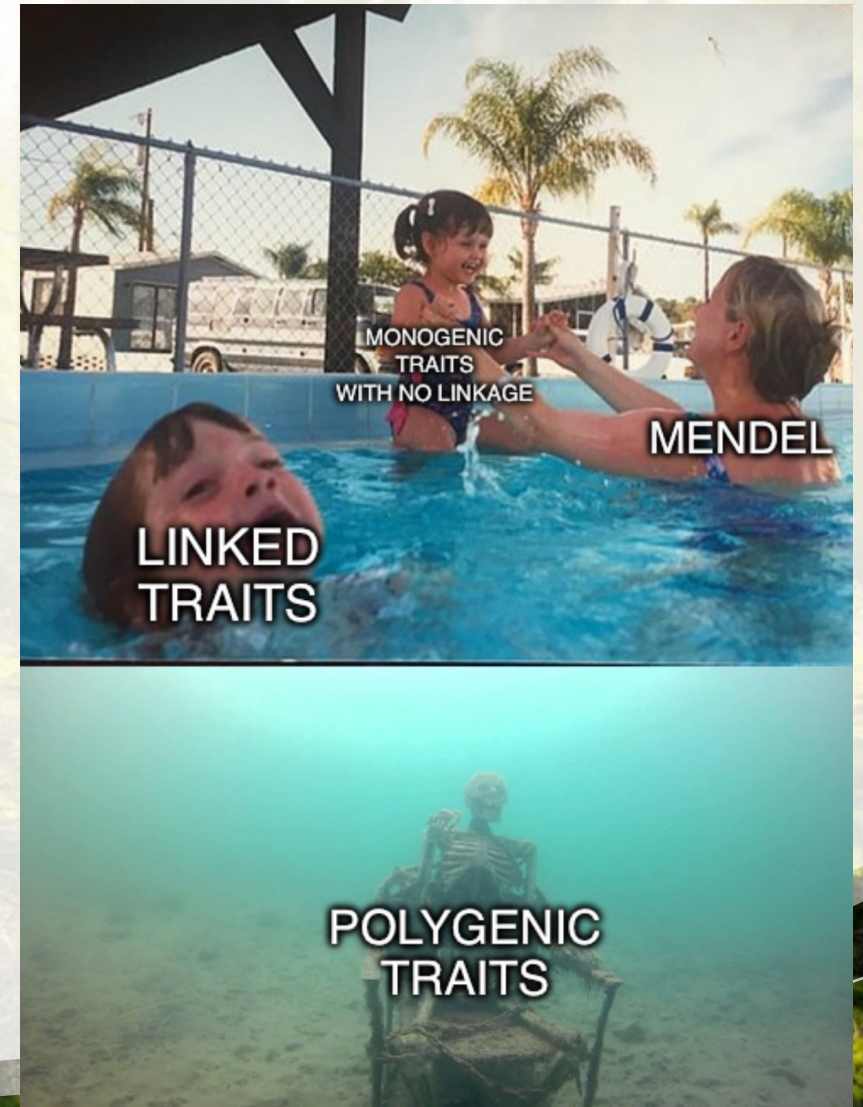
Learning goals:

The causes and consequences of non-dominant inheritance.

How gene interactions leads to epistasis.

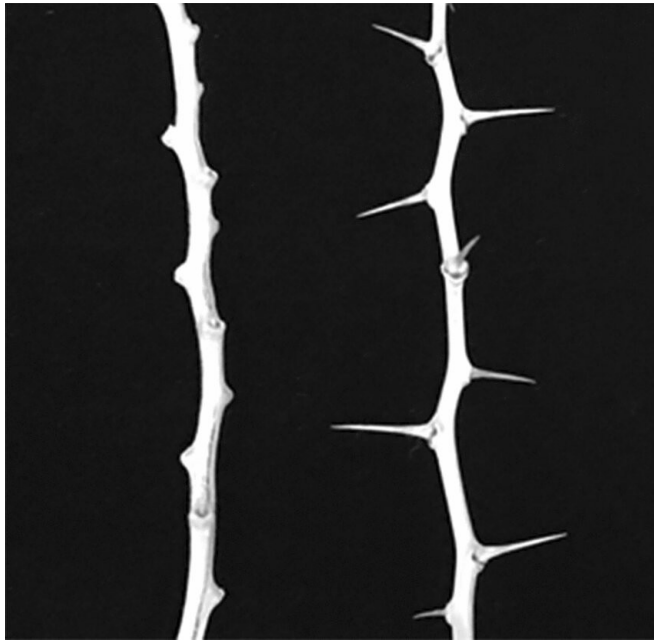
How linkage on the same chromosome violates independent assortment.

...

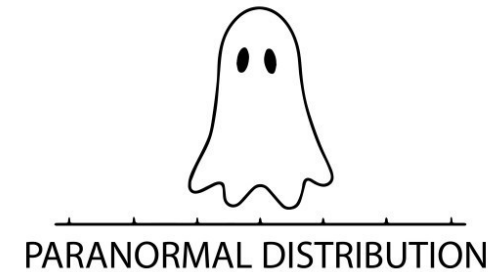
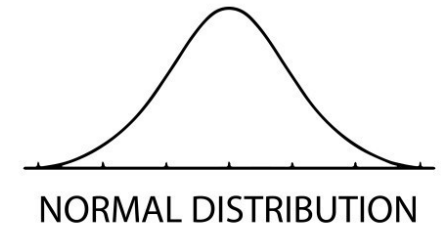
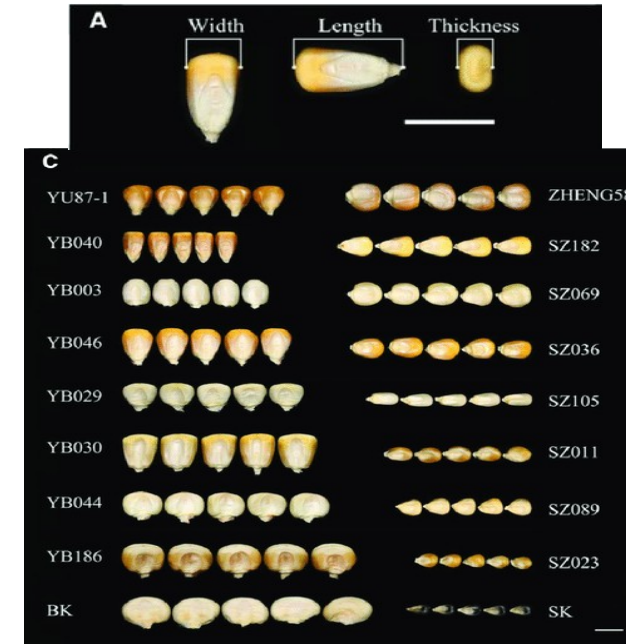


Qualitative vs quantitative traits

Qualitative (e.g. thorns absent vs present)

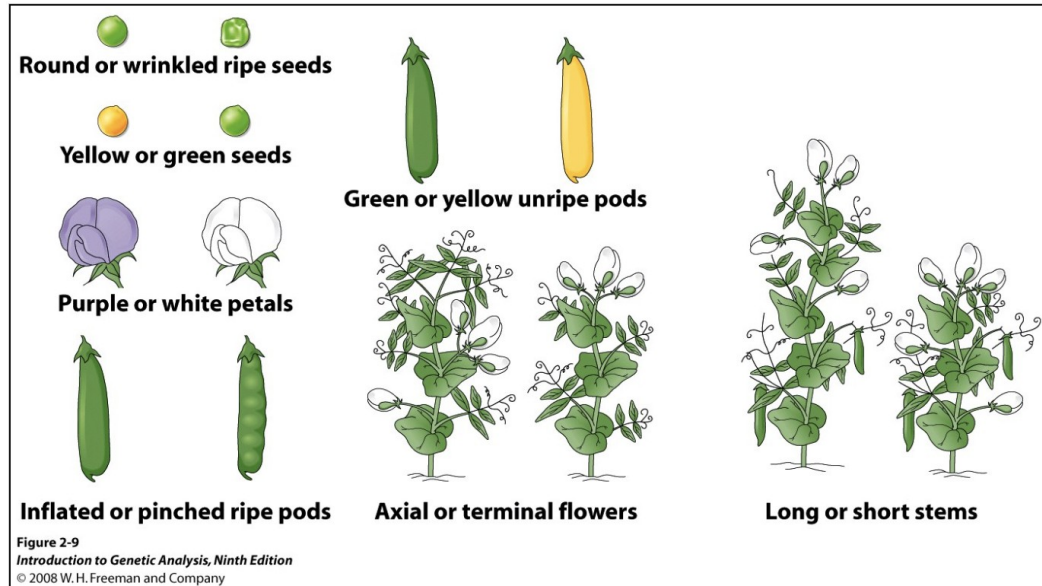


Quantitative (e.g. Kernel length)

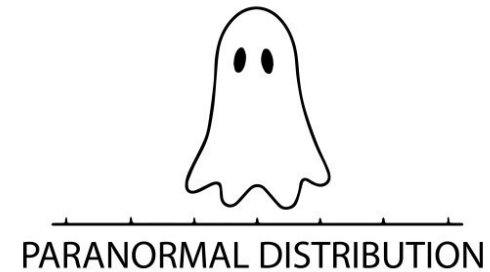
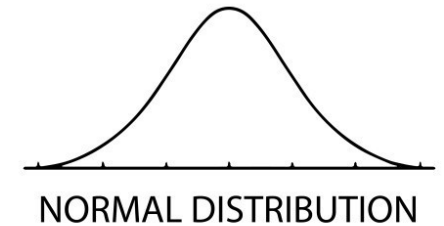
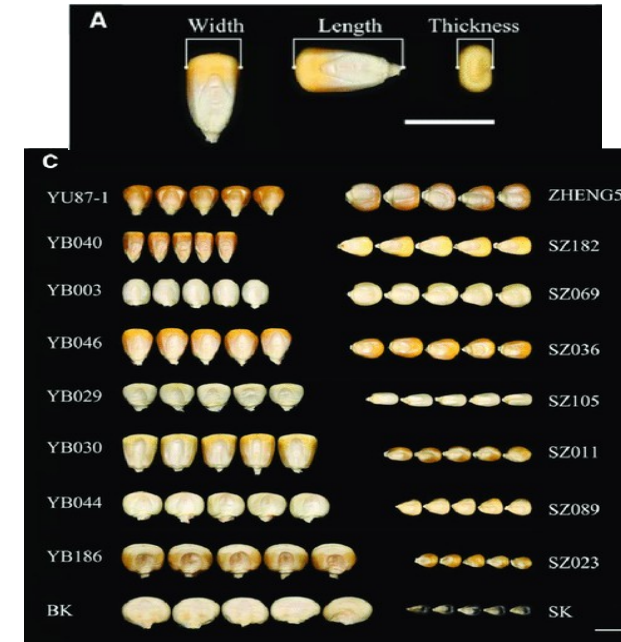


Qualitative vs quantitative traits

Qualitative (e.g. thorns absent vs present)

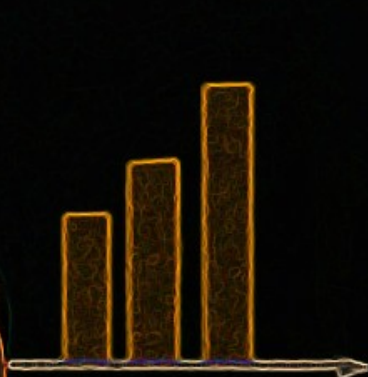


Quantitative (e.g. Kernel length)



MENDELIANS

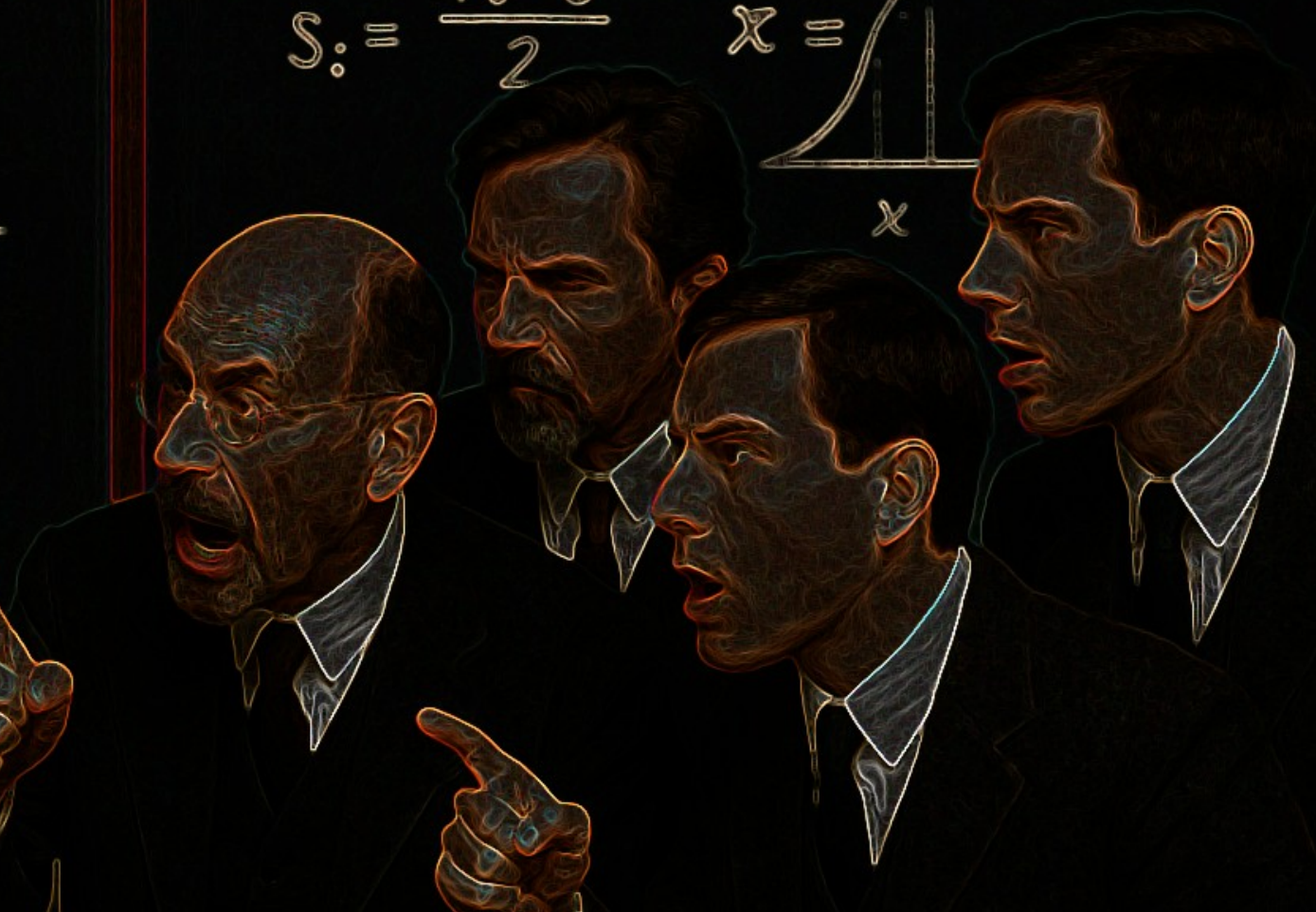
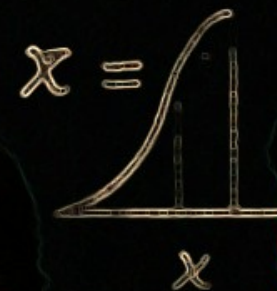
AA	a
A+	



BIOMETRICIANS

$$y = a + bx$$

$$s := \frac{n-6}{2}$$



MENDELIANS

Conflict?

BIOMETRICIANS

Qualitative “Mendelian” trait

Parent 1

Parent 2

X
↓

Quantitative trait

Live poll

Word cloud

Think of as many as you can: traits you could measure in a plant



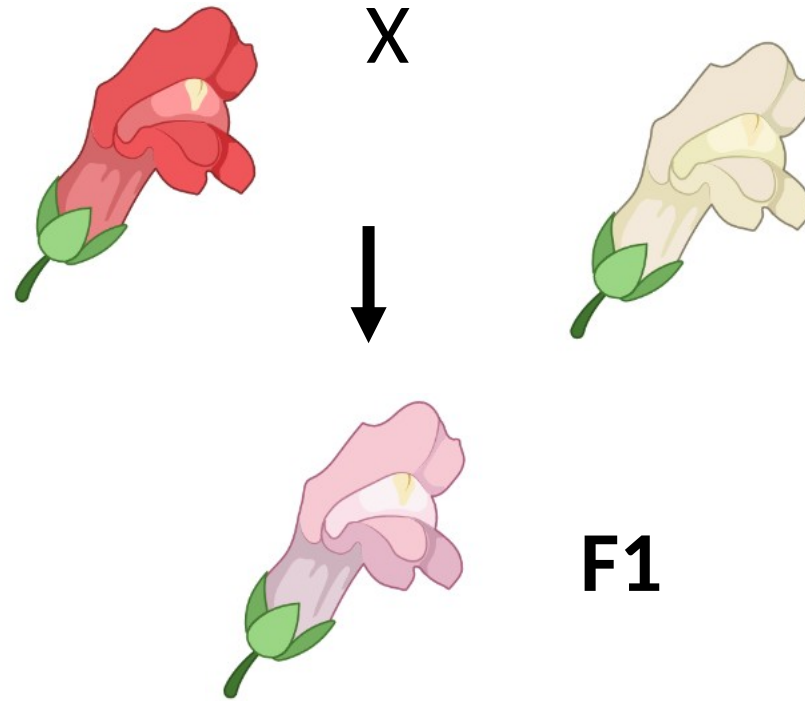
Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-traits](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-traits)

[host screen](#)

Incomplete dominance

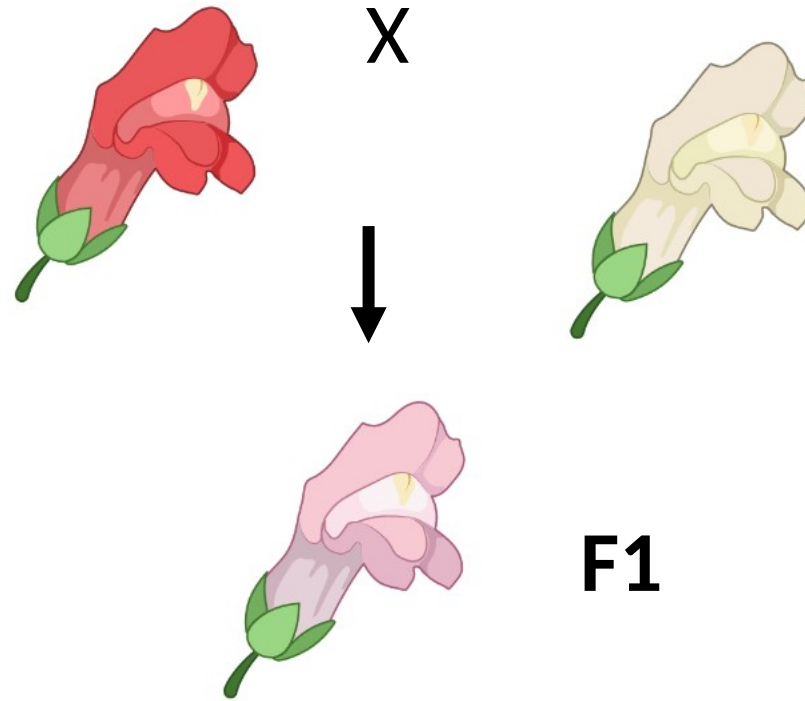
Snapdragons



Incomplete dominance



Blending inheritance?!



Snapdragons

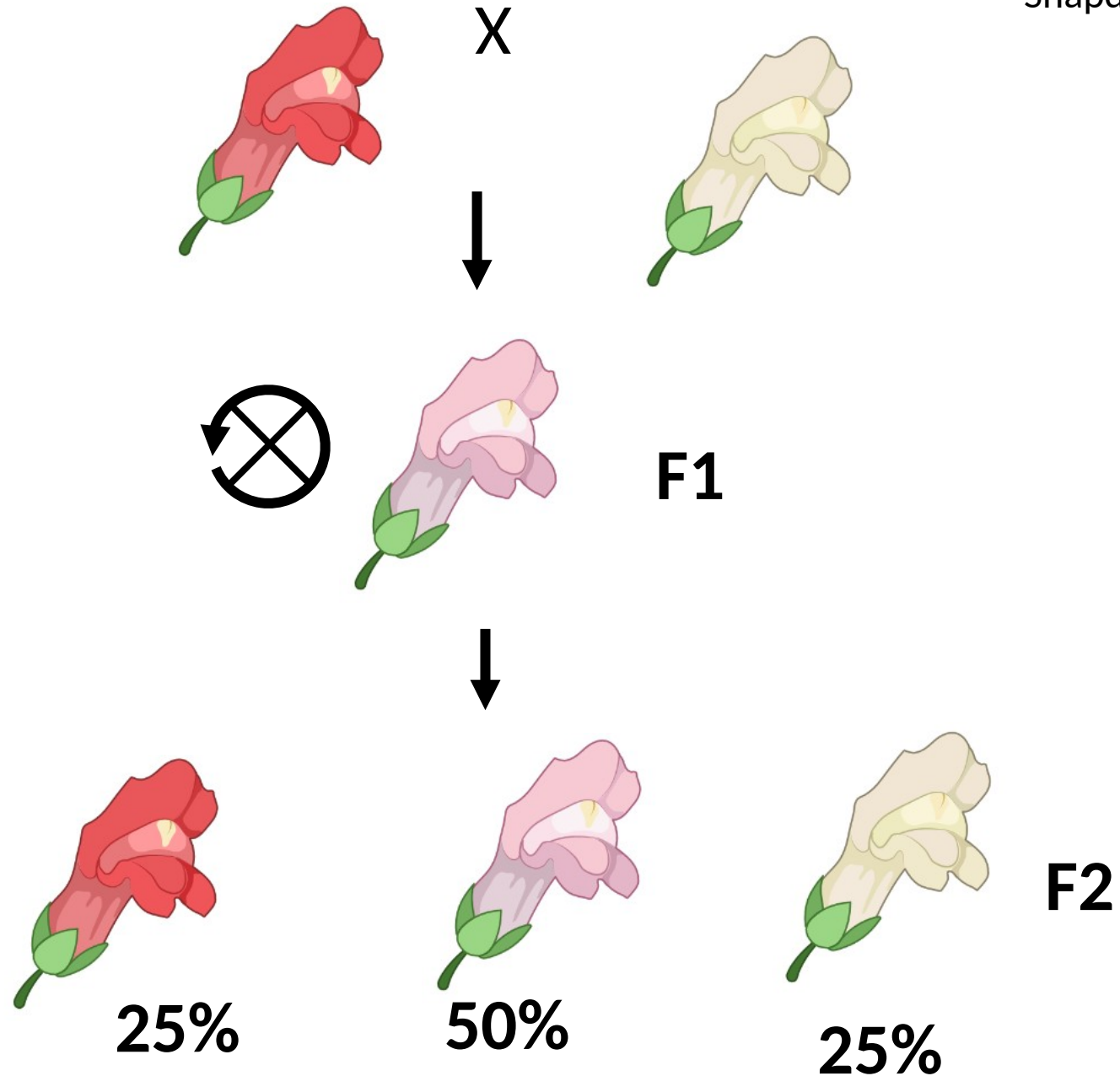
Incomplete dominance

Snapdragons



Blending inheritance?!

Why is this
different from
blending
inheritance?



Live poll

Text wall

Why is incomplete dominance different from blending inheritance?



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-notblending](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-notblending)

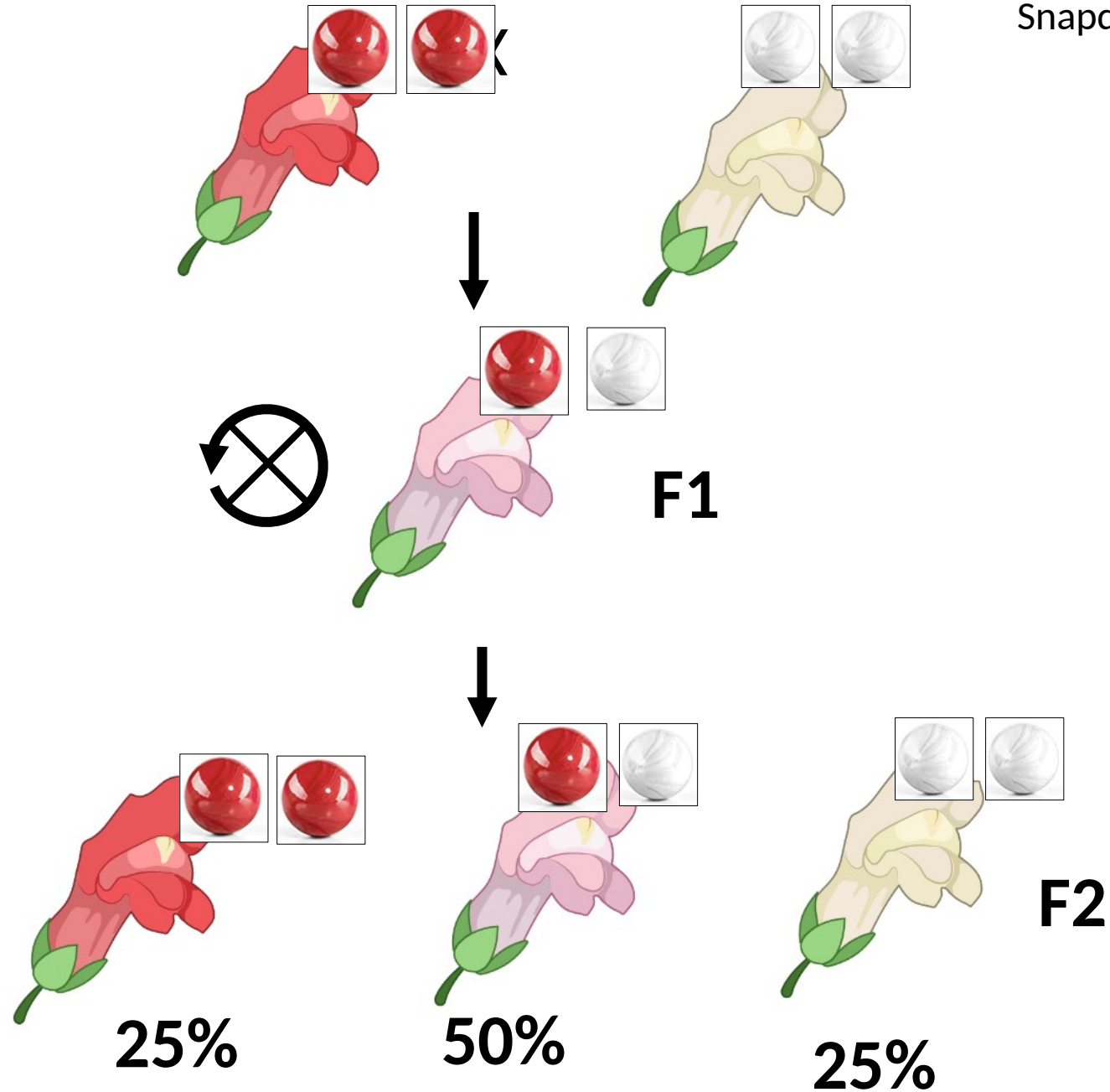
[host screen](#)

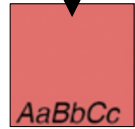
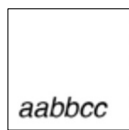
Incomplete dominance



Why is this
different from
blending
inheritance?

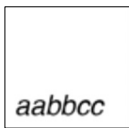
Snapdragons





Each uppercase allele (A,B,C) contributes 1 unit of pigment

Incomplete dominance in many genes leads to “additive genetic variation”



Each uppercase allele (A,B,C) contributes 1 unit of pigment

Incomplete dominance in many genes leads to “additive genetic variation”

	ABC	AbC	aBC	ABc	Abc	abC	aBc	abc
ABC	AABBCC	AABbCC	AaBBCC	AABBCc	AABbCc	AaBbCC	AaBBCc	AaBbCc
AbC	AABbCC	AAbbCC	AaBbCC	AABbCc	AAbbCc	AabbCC	AaBbCc	AabbCc
aBC	AaBBCC	AaBbCC	aaBBCC	AaBBCc	AaBbCc	aaBbCC	aaBBCc	aaBbCc
ABc	AABBCc	AABbCc	AaBBCc	AABBcc	AABbcc	AaBbCc	AaBBcc	AaBbcc
Abc	AABbCc	AAbbCc	AaBbCc	AABbcc	AAbbcc	AabbCc	AaBbcc	Aabbcc
abC	AaBbCC	AabbCC	aaBbCC	AaBbCc	AabbCc	aabbCC	aaBbCc	aabbCc
aBc	AaBBCc	AaBbCc	aaBBCc	AaBBcc	AaBbcc	aaBbCc	aaBBcc	aaBbcc
abc	AaBbCc	AabbCc	aaBbCc	AaBbcc	Aabbcc	aabbCc	aaBbcc	aabbcc

AABBCC

aabbcc

AaBbCc

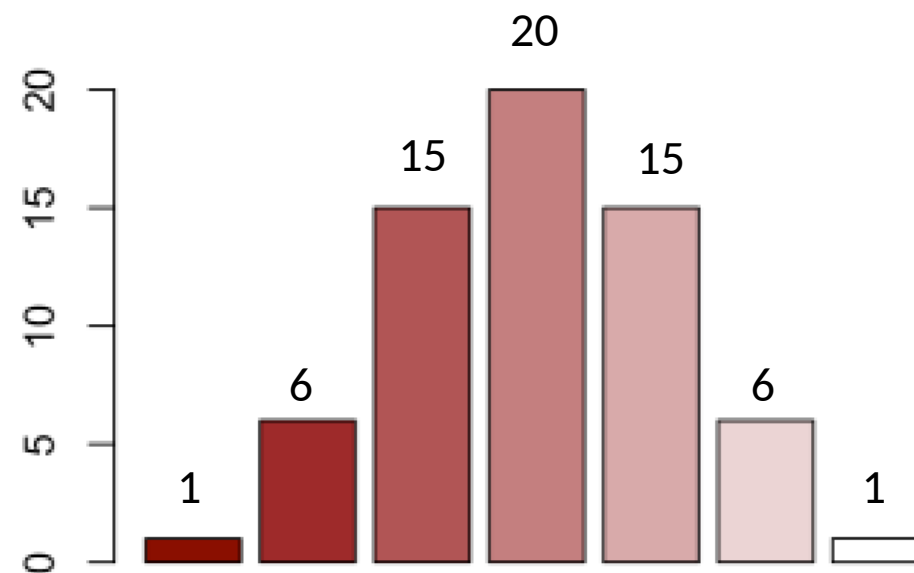


Each uppercase allele (A,B,C) contributes 1 unit of pigment

Incomplete dominance in many genes leads to “additive genetic variation”

	ABC	AbC	aBC	ABc	Abc	abC	aBc	abc
ABC	AABBCC	AABbCC	AaBBCC	AABBCc	AABbCc	AaBbCC	AaBBCc	AaBbCc
AbC	AABbCC	AAbbCC	AaBbCC	AABbCc	AAbbCc	AabbCC	AaBbCc	AabbCc
aBC	AaBBCC	AaBbCC	aaBBCC	AaBBCc	AaBbCc	aaBbCC	aaBBCc	aaBbCc
ABc	AABBCc	AABbCc	AaBBCc	AABBcc	AABbcc	AaBbCc	AaBBcc	AaBbcc
Abc	AABbCc	AAbbCc	AaBbCc	AABbcc	AAbbcc	AabbCc	AaBbcc	Aabbcc
abC	AaBbCC	AabbCC	aaBbCC	AaBbCc	AabbCc	aabbCC	aaBbCc	aabbCc
aBc	AaBBCc	AaBbCc	aaBBCc	AaBBcc	AaBbcc	aaBbCc	aaBBcc	aaBbcc
abc	AaBbCc	AabbCc	aaBbCc	AaBbcc	Aabbcc	aabbCc	aaBbcc	aabbcc

Frequency



Phenotype

possible phenotypes = # genes * 2 + 1

of Genes: 1



of Genes: 1



of Genes: 2



of Genes: 1



of Genes: 2



of Genes: 3



of Genes: 1



of Genes: 2



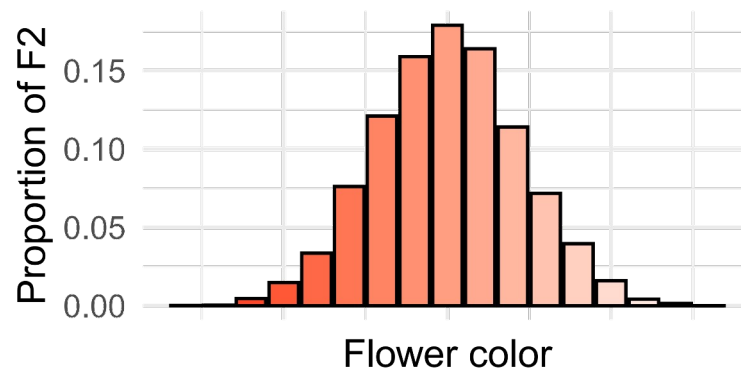
of Genes: 3



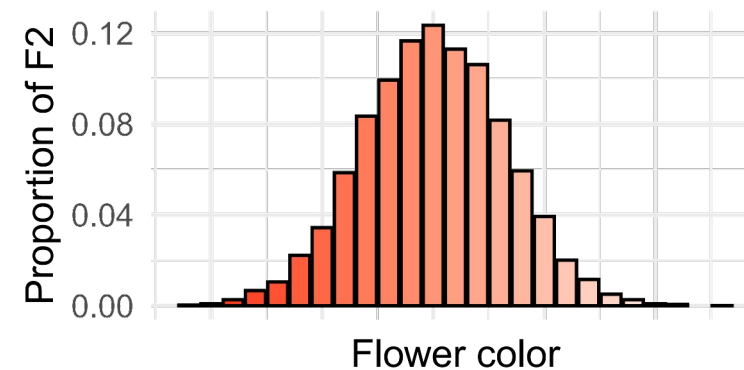
of Genes: 5



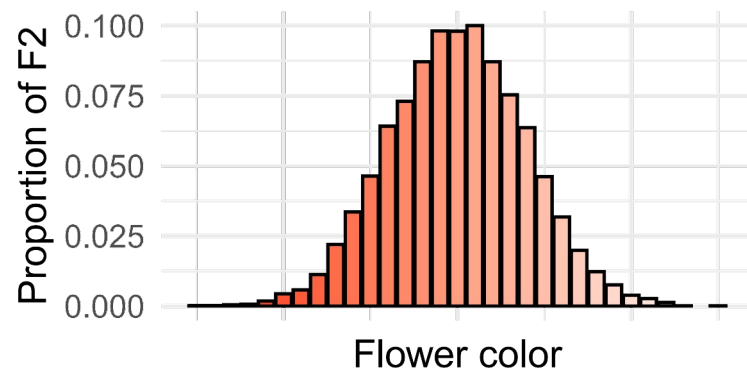
of Genes: 10



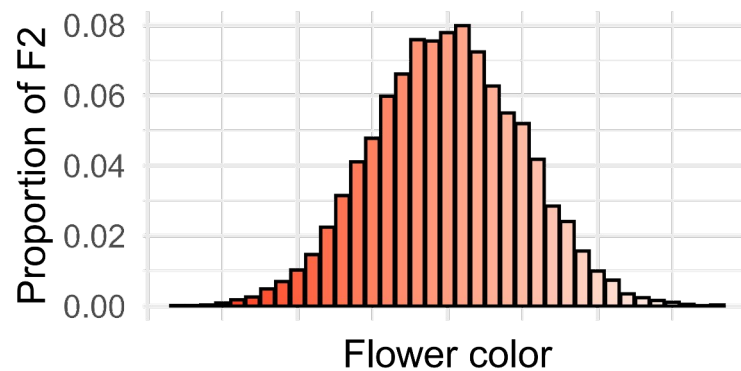
of Genes: 20



of Genes: 30



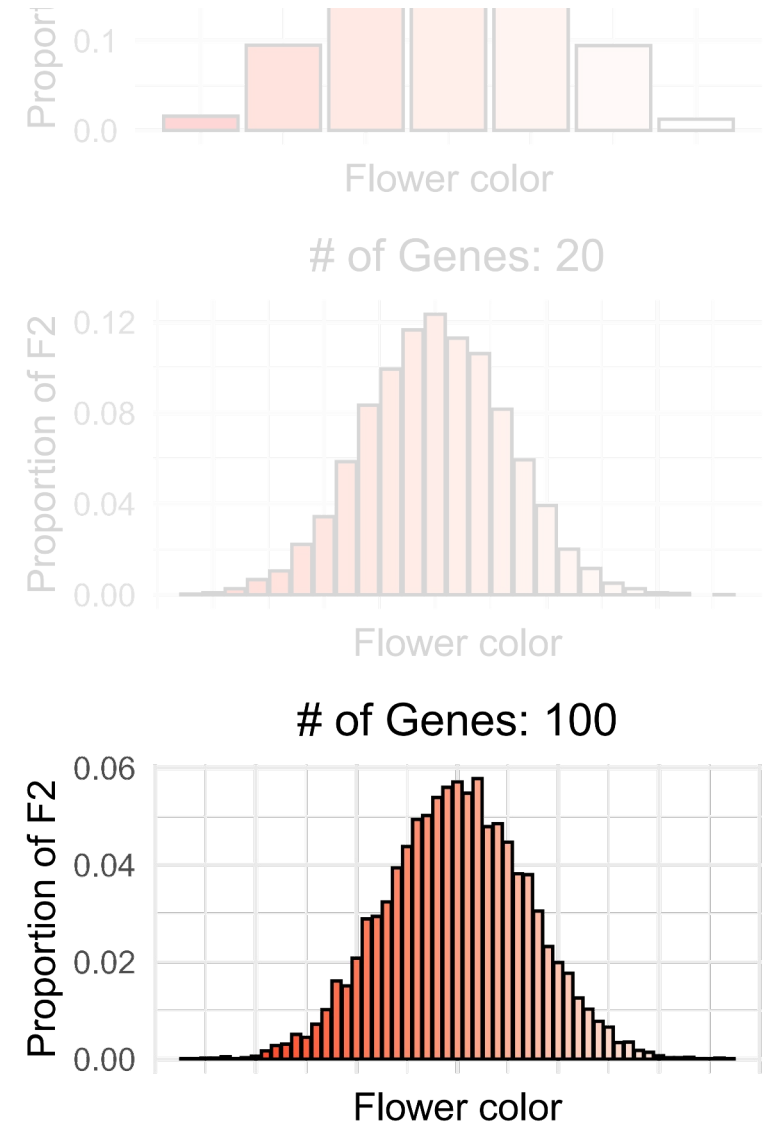
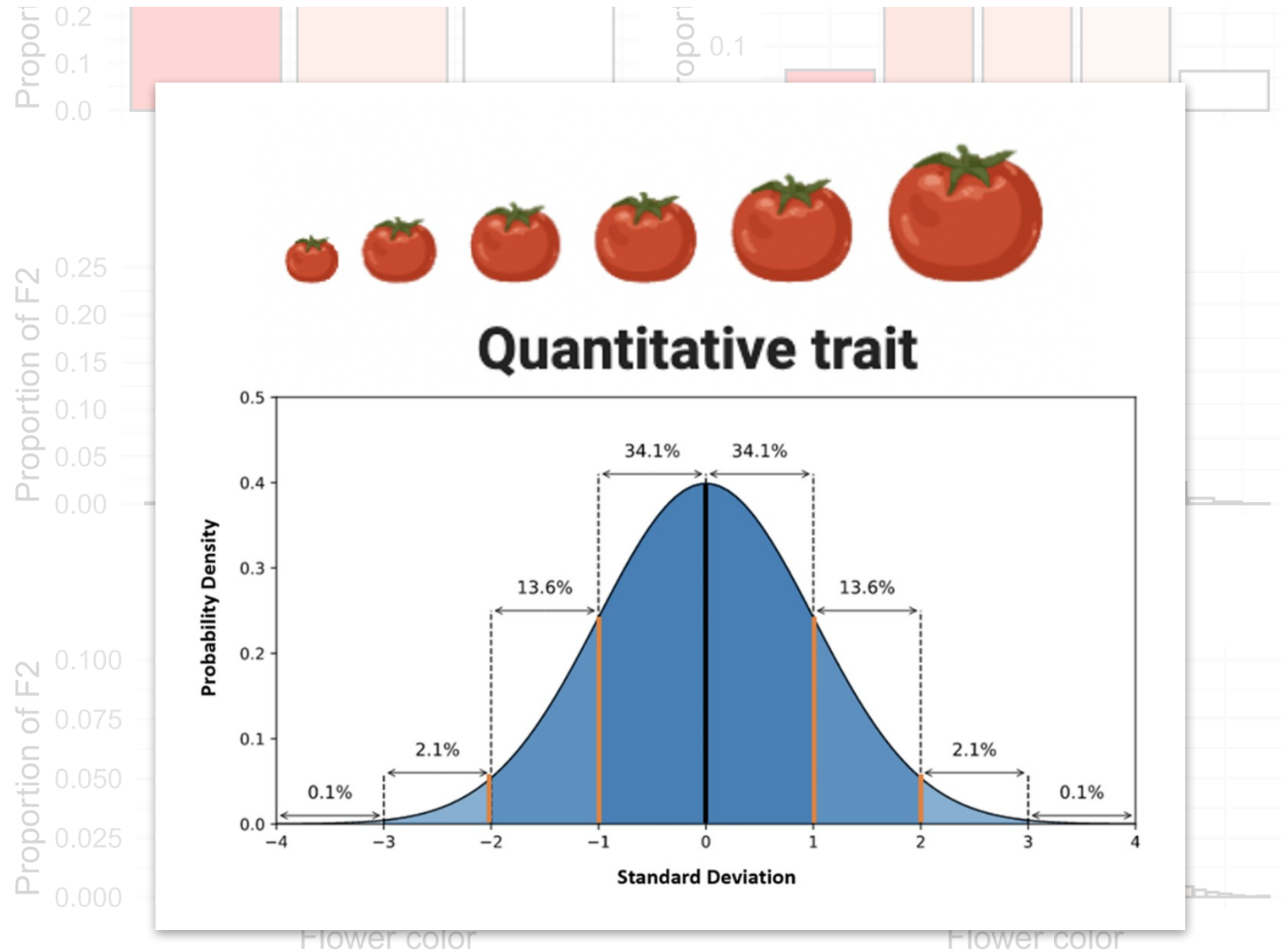
of Genes: 50



of Genes: 100



Quantitative traits are consistent with Mendelian genetics if polygenic: many genes contribute to the trait



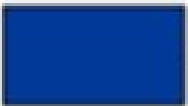
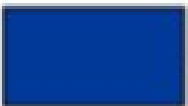
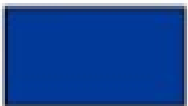
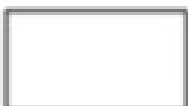
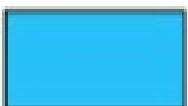
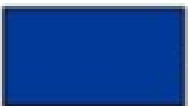
“Quantitative Genetics”

will cover more later this quarter but for now...

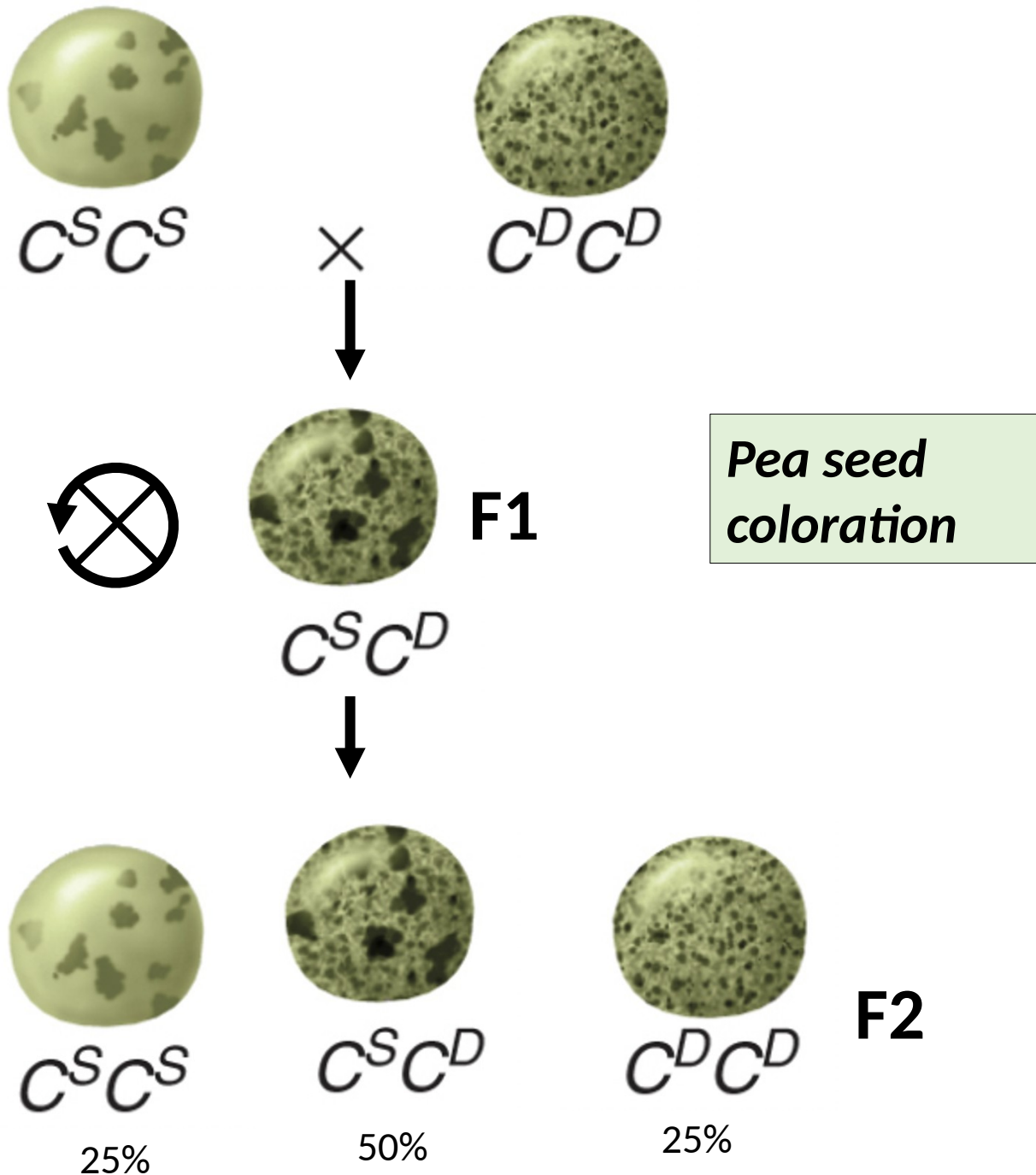
Phenotype = Genotype + Environment

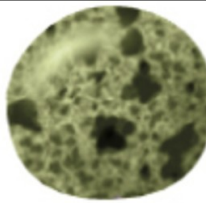
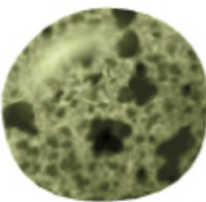
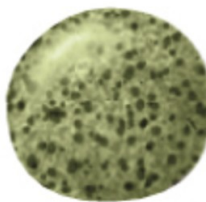
Summary: types of dominance

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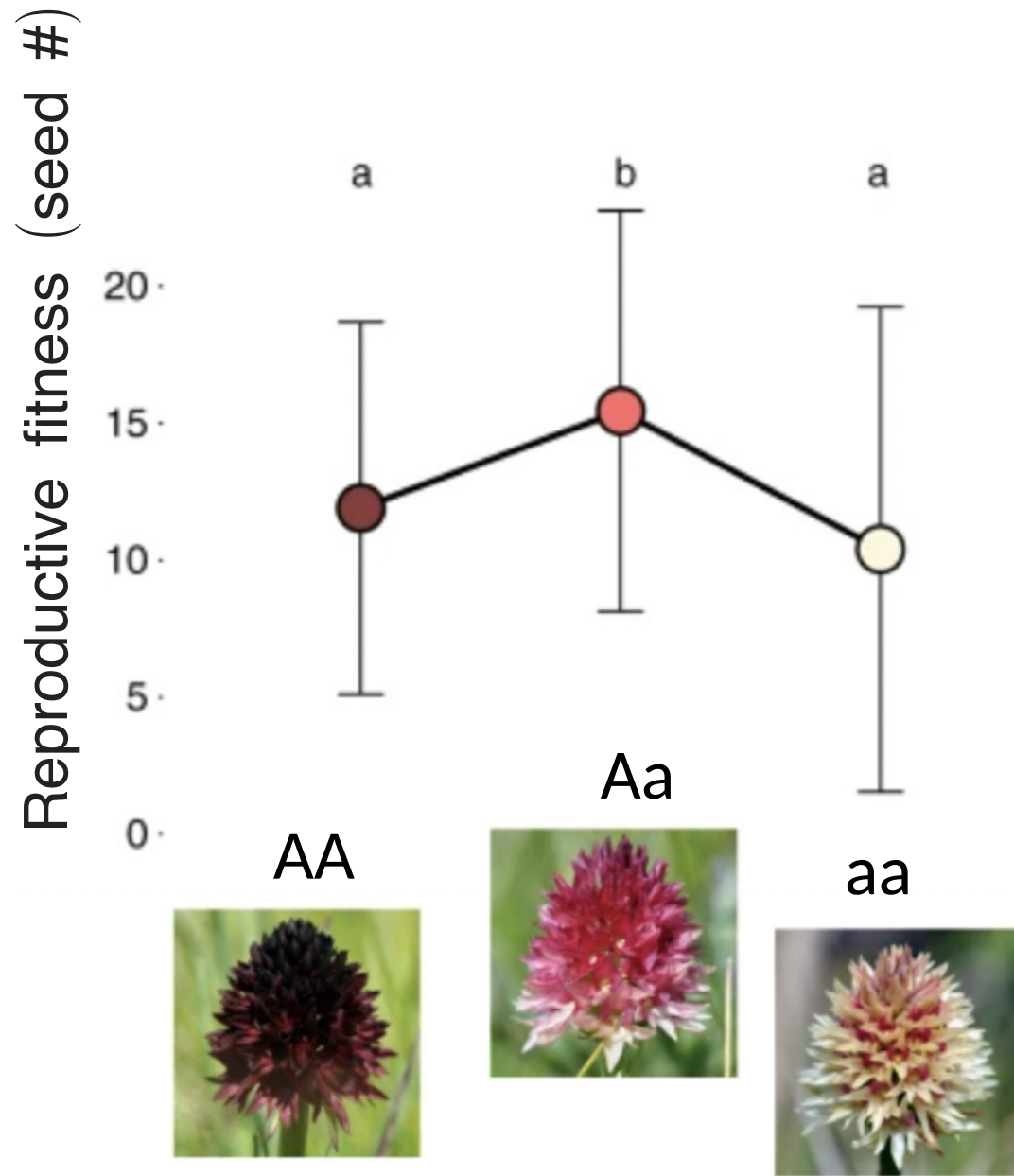
Type of dominance	A^1/A^1	A^2/A^2	A^1/A^2 hybrids	
Complete				A^1 is dominant to A^2 A^2 is recessive to A^1
Complete				A^2 is dominant to A^1 A^1 is recessive to A^2
Incomplete				A^1 and A^2 are incompletely dominant relative to each other
Codominant				A^1 and A^2 are codominant relative to each other

Co-dominance: both alleles show phenotype



	C^S	C^D
C^S	 $C^S C^S$	 $C^S C^D$
C^D	 $C^S C^D$	 $C^D C^D$

Overdominance:
best of both worlds aka
“heterozygote advantage”



**Why might the
heterozygote
have better
fitness?**

Live poll

Text wall

Why might the heterozygote have better fitness?

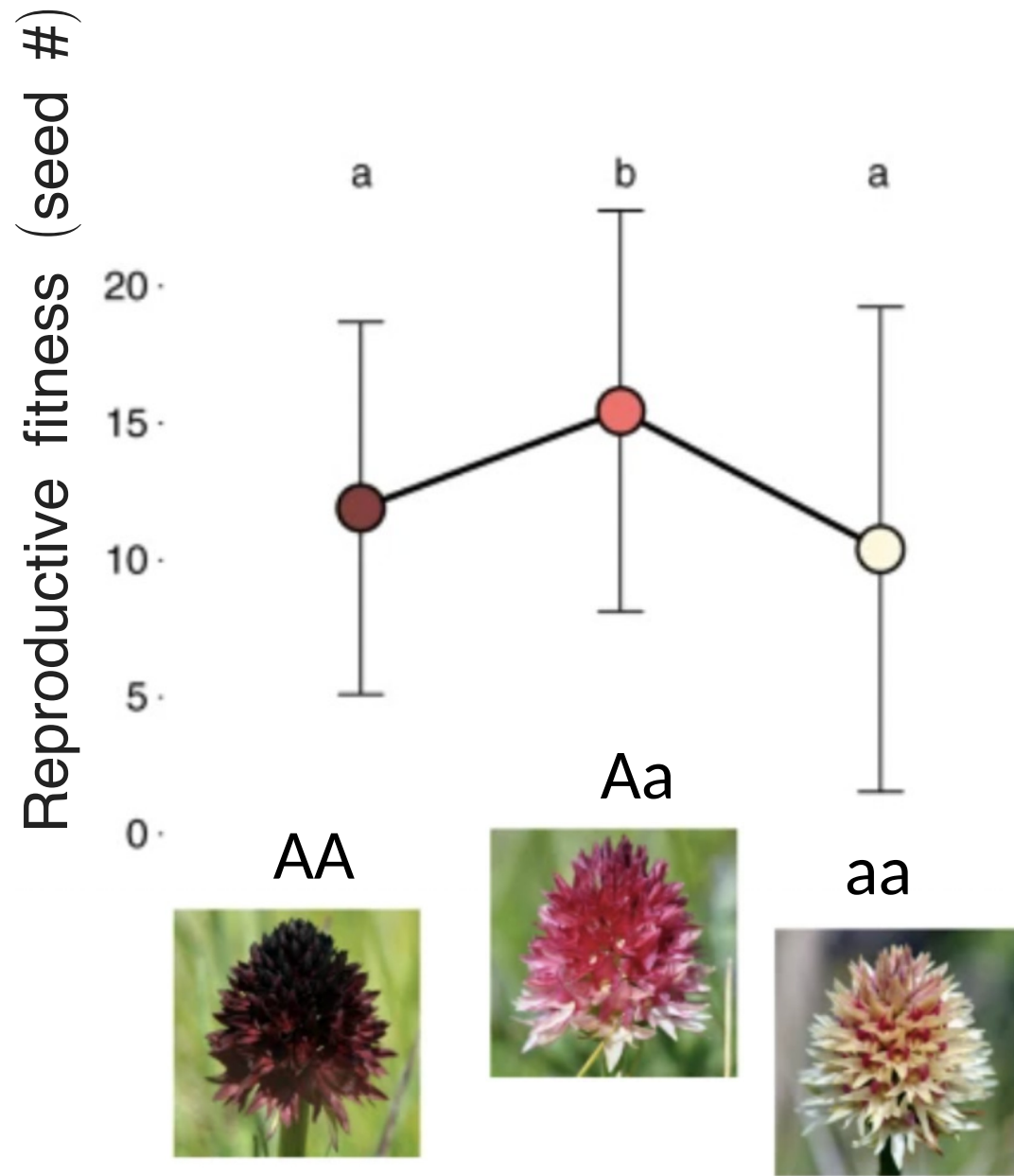


Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-overdom](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-overdom)

[host screen](#)

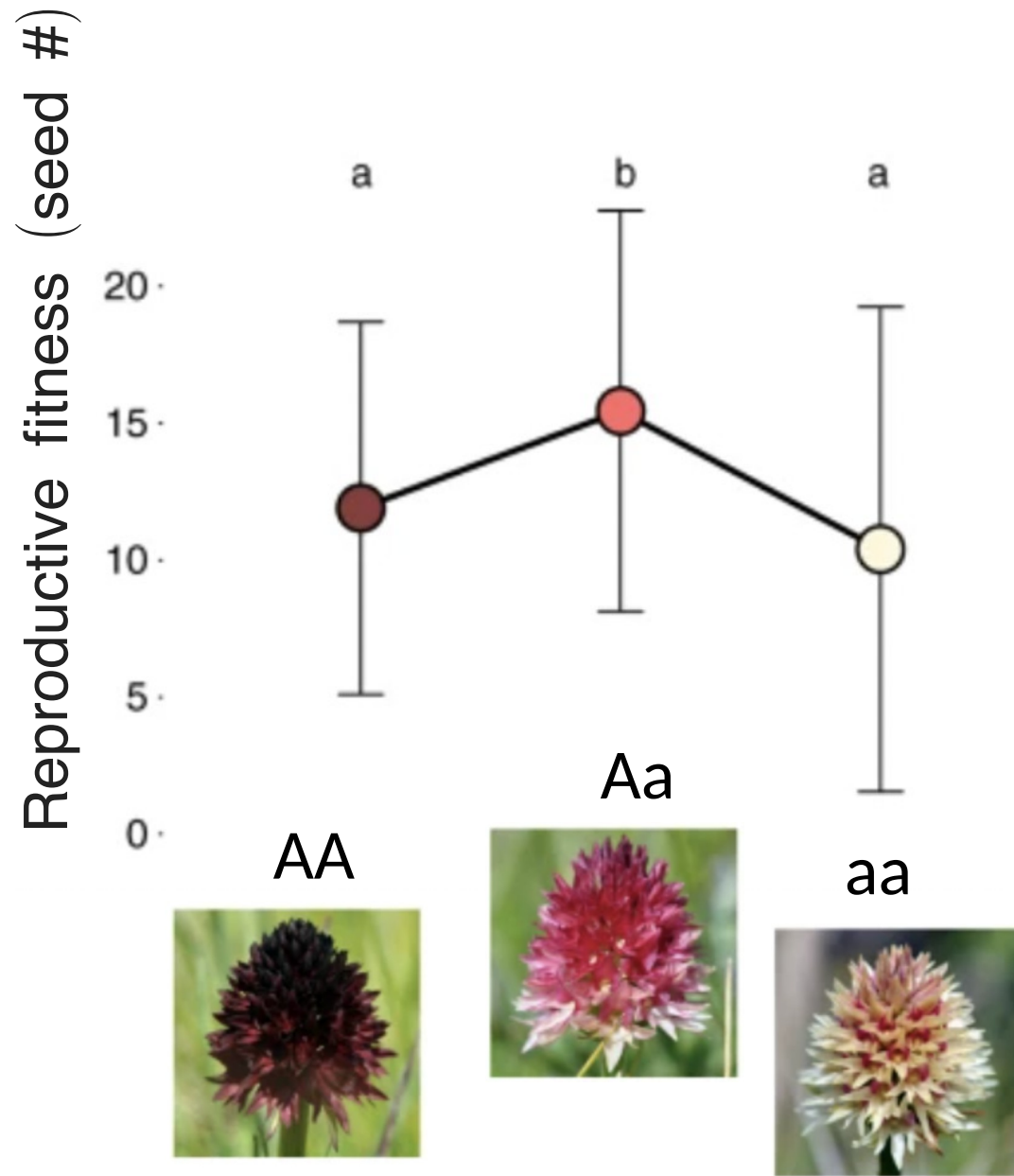
Overdominance:
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**Why might the
heterozygote
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Alpine orchid *Gymnadenia
rhellicani*

Overdominance:
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**Why might the
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Alpine orchid *Gymnadenia
rhellicani*

Disease resistance A/A
 b/b



X
↓



a/a
 B/B Ear size

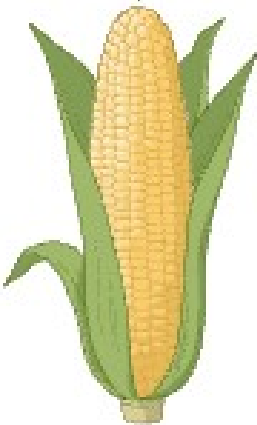


F1
Hybrid – seed sold to farmers



↓

A/a
 b/B



Disease resistance A/A
 b/b



X
↓



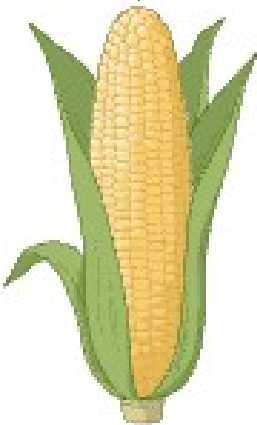
a/a
 B/B Ear size



A/a
 b/B



↓



F1
Hybrid – seed sold to farmers



HYBRID VIGOR

Disease resistance A/A
 b/b



X
↓



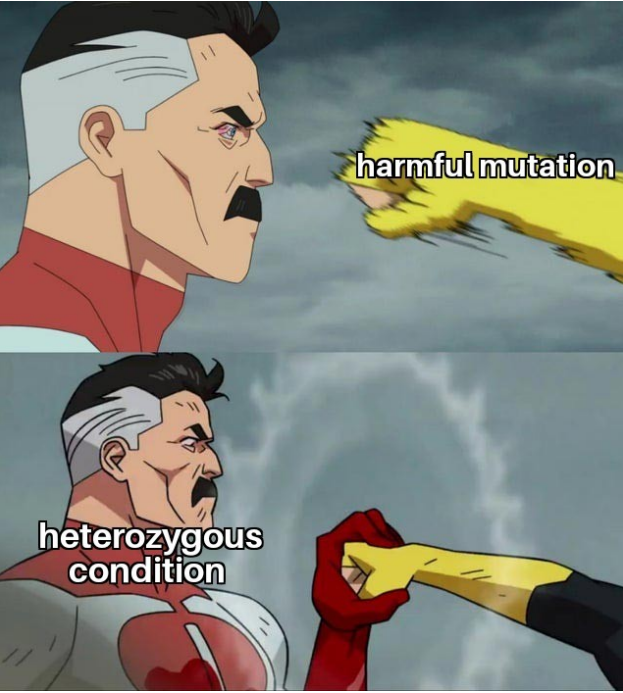
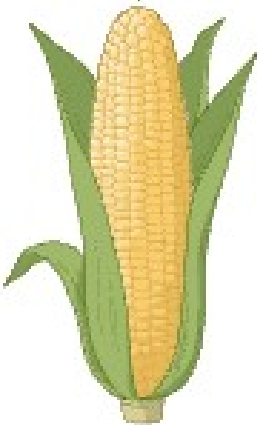
a/a
 B/B Ear size



F1
Hybrid – seed sold to farmers

↓

A/a
 b/B



Disease resistance A/A
 b/b



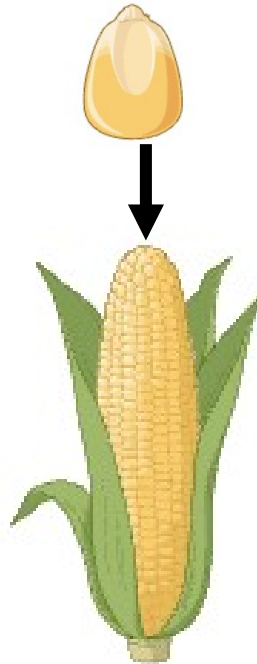
X
↓



a/a
 B/B Ear size



A/a
 b/B



F1

Hybrid – seed sold to farmers

F1 Hybrids

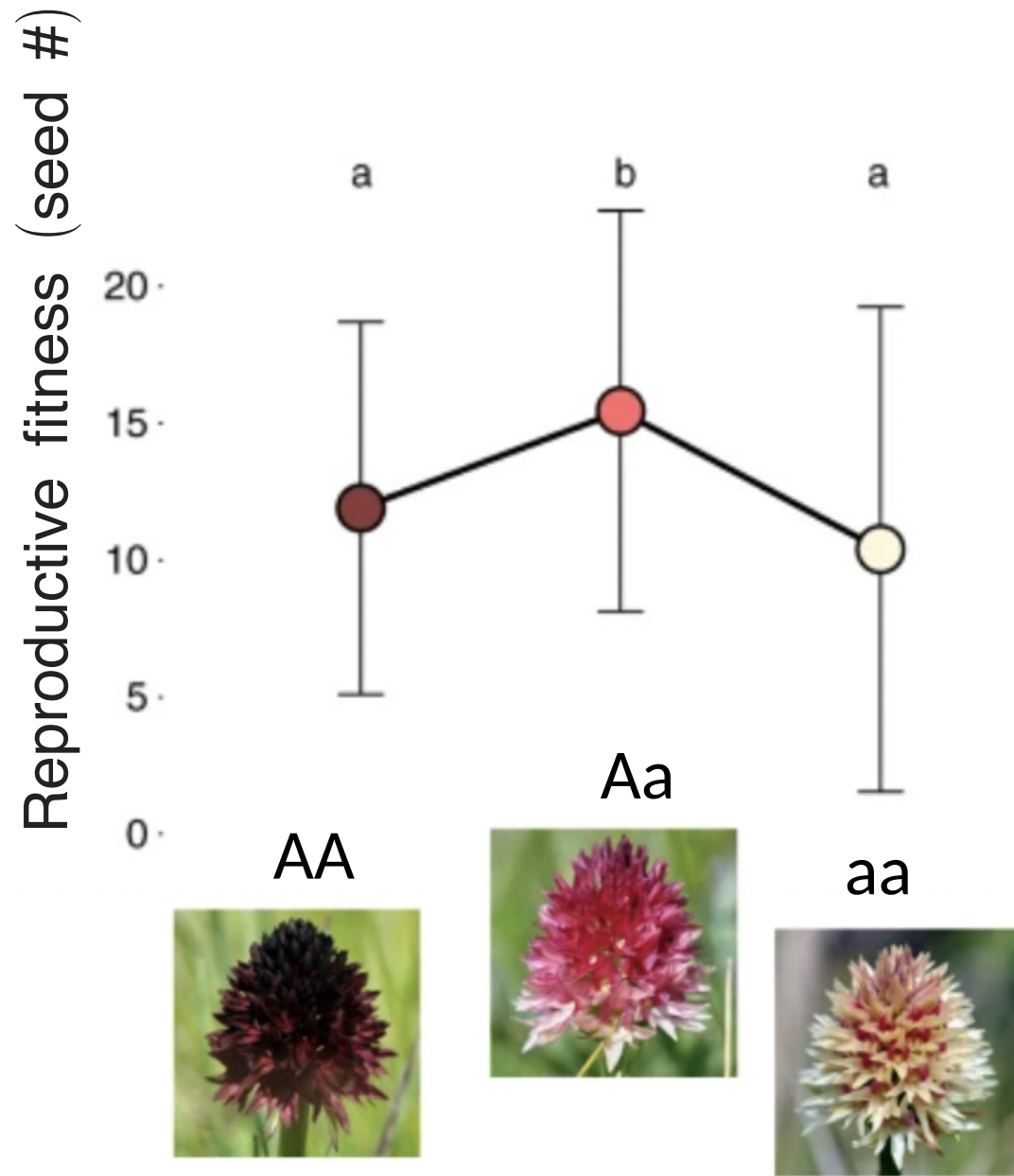


F2



F2 will have some homozygous deleterious alleles and be less genetically uniform

Overdominance:
best of both worlds aka
“heterozygote advantage”



**Why might the
heterozygote
have better
fitness?**

Alpine orchid *Gymnadenia
rhellicani*

Heterosis (AKA hybrid vigor): overdominance?



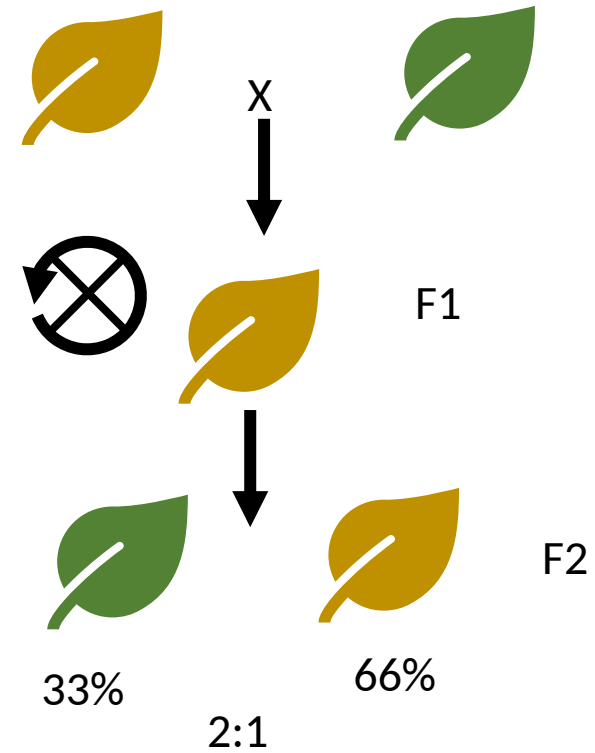
A photograph of a forest floor with sunlight filtering through the trees. A semi-transparent white circular overlay is positioned in the upper half of the image, containing the text 'Other deviations from Mendelian expectations...'.

Other deviations from Mendelian expectations...



Nature Reviews | **Genetics**

Snapdragon



YG x YG

	Y	G
Y	YY	YG
G	YG	GG

Why might we see a 2:1 ratio in the F2?

Live poll

Text wall

Why might we see a 2:1 ratio in the F2?



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-twoone](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-twoone)

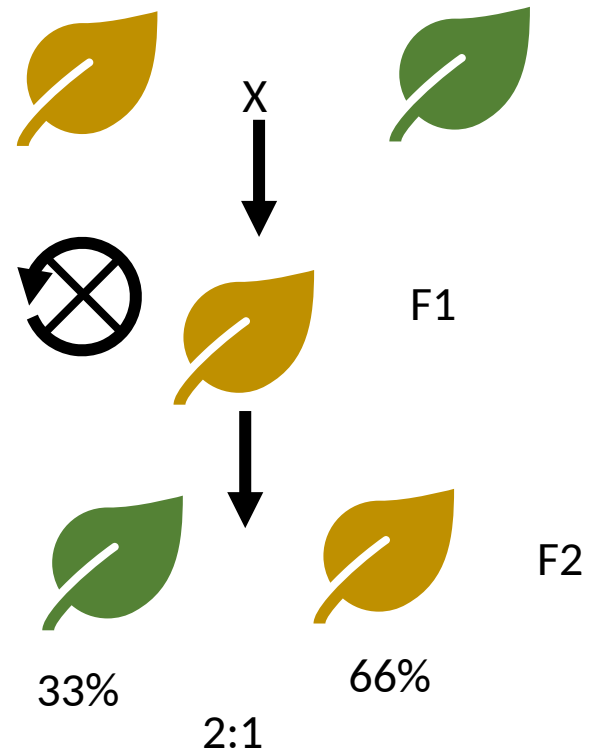
[host screen](#)

Homozygous lethal alleles



Nature Reviews | Genetics

Snapdragon



YG x YG

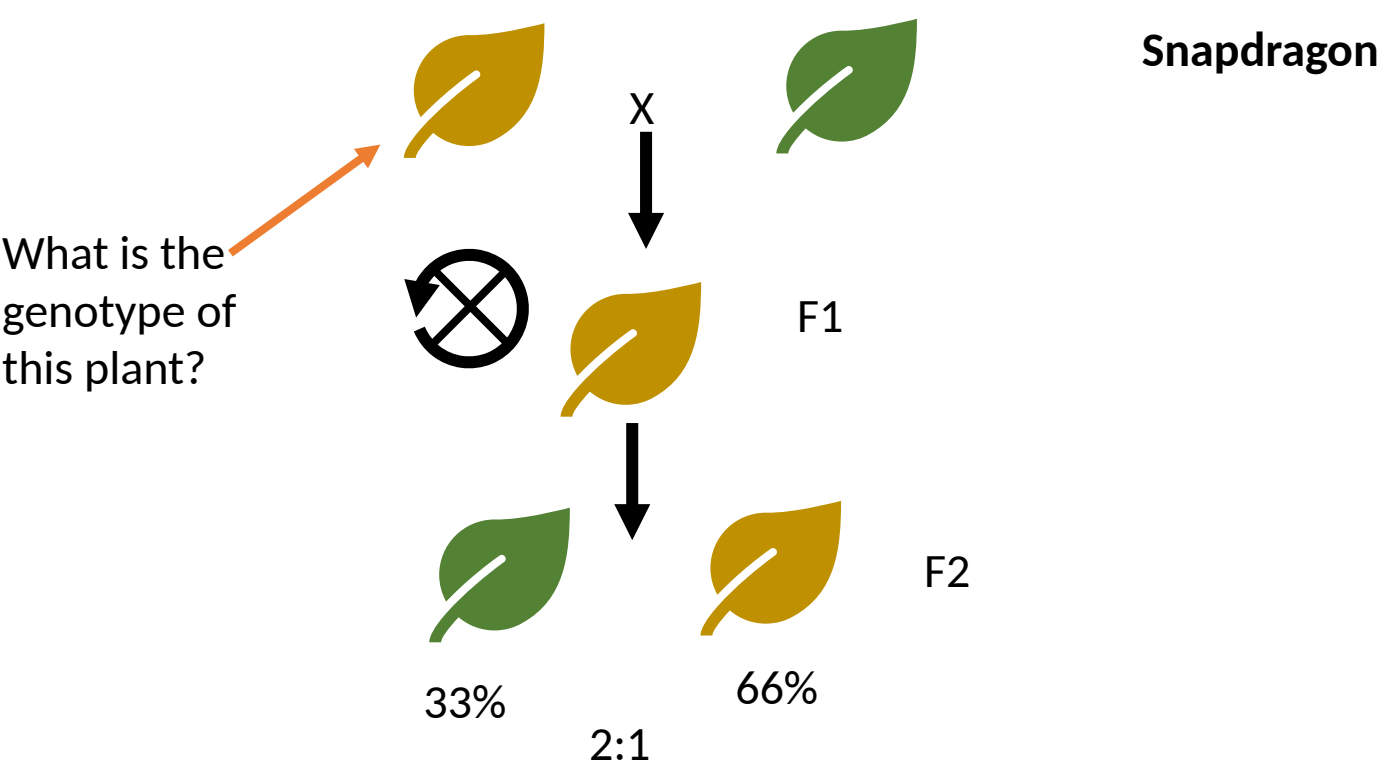
	Y	G	
Y	YY	YG	
G	YG	GG	

2:1

Homozygous lethal alleles



Nature Reviews | Genetics



YG x YG

	Y	G
Y	YY	YG
G	YG	GG

2:1

Live poll

Multiple choice

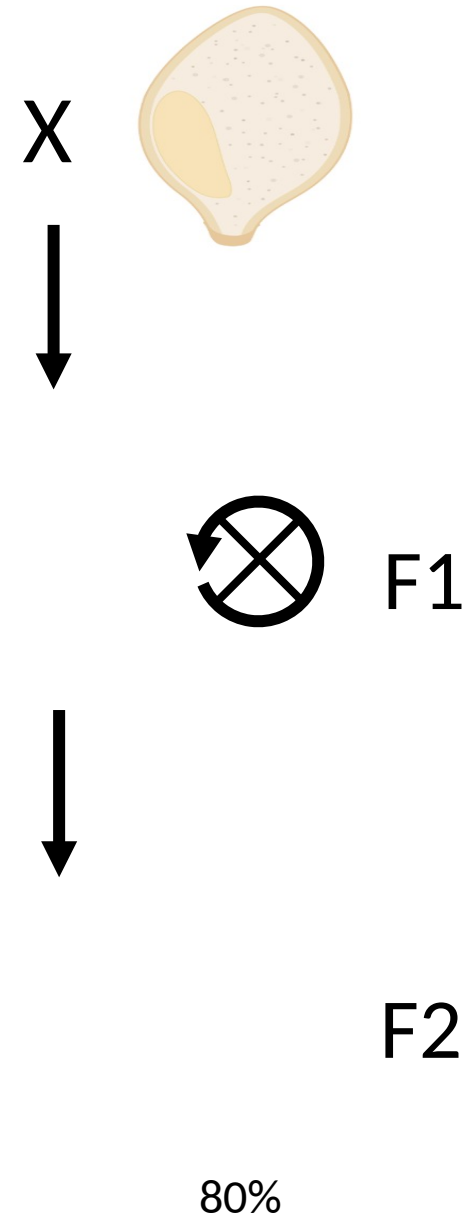
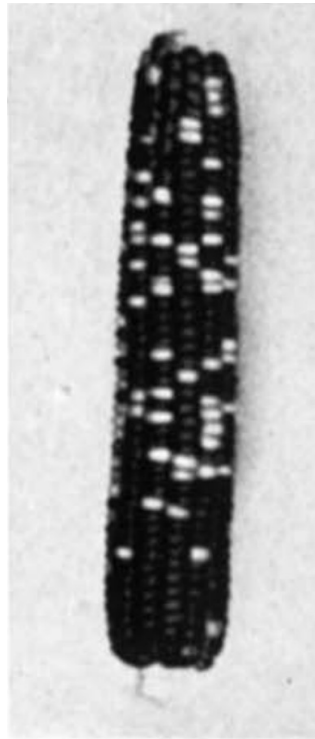
YG × YG snapdragons: what is the genotype of this yellow-green plant?



Scan with your phone

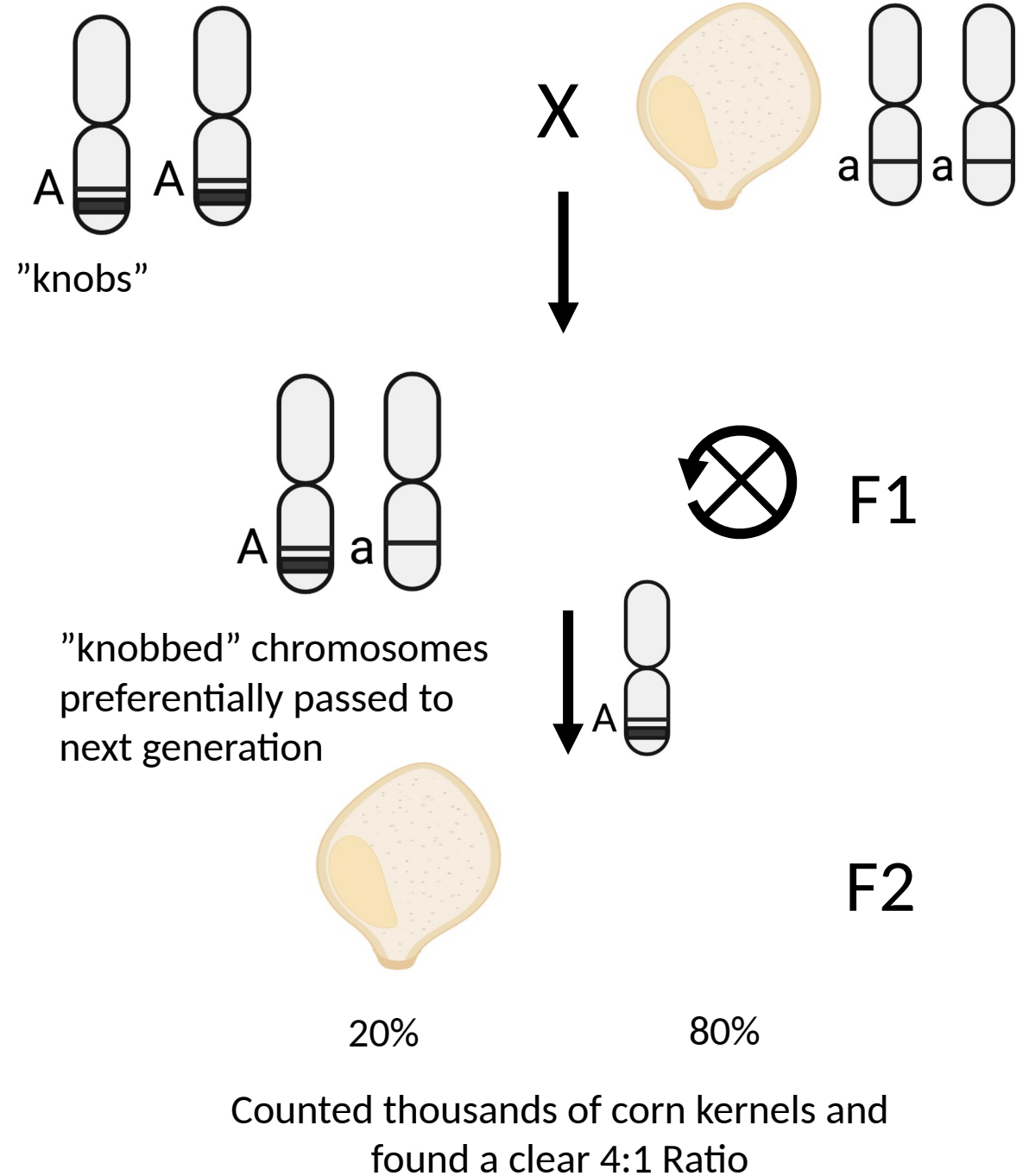
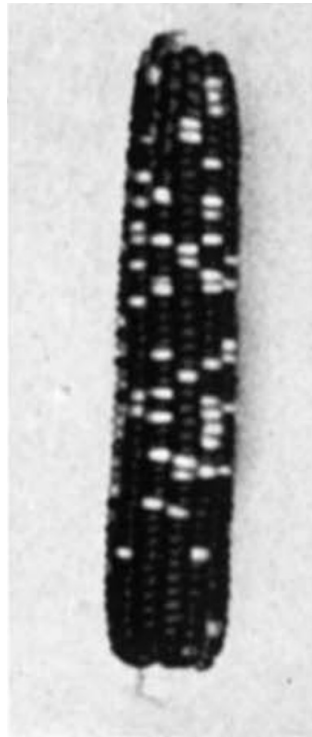
[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-lethalgen](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-lethalgen)
[o](#)

[host screen](#)

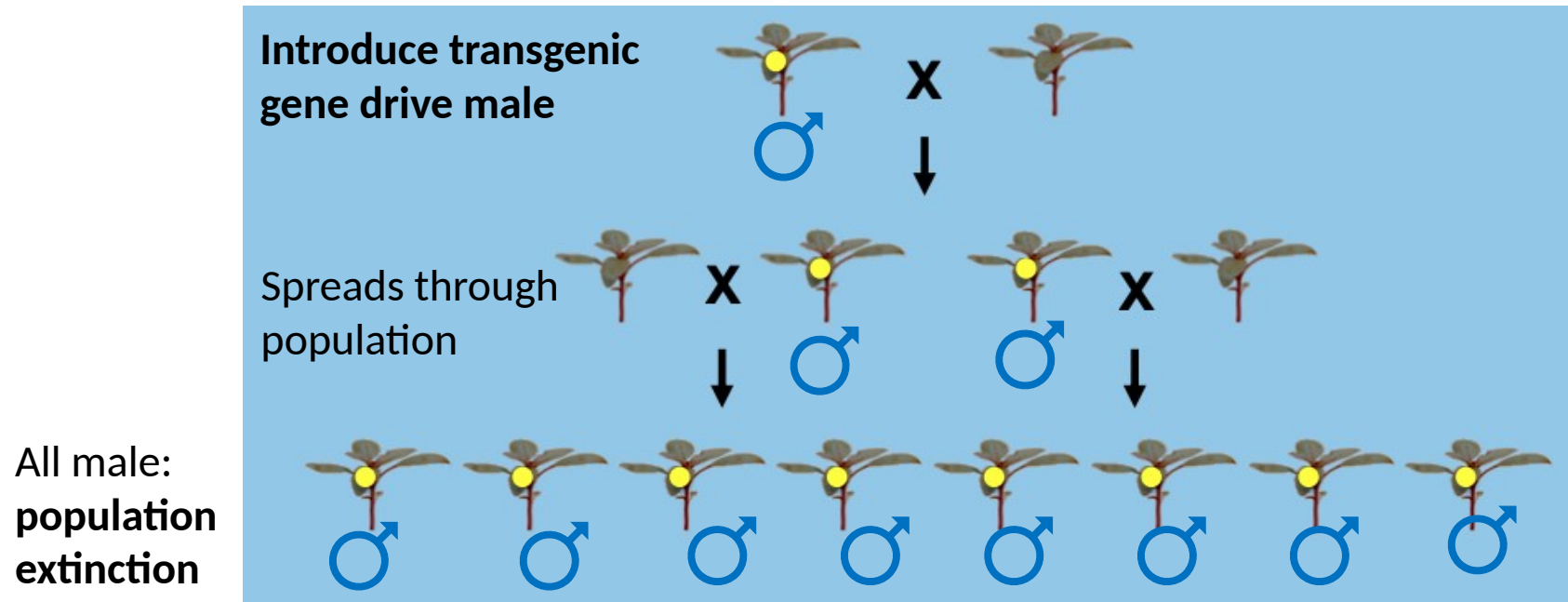


Counted thousands of corn kernels and
found a clear 4:1 Ratio

Meiotic drive: one allele inherited more often than another



Should we use meiotic drive to eliminate weeds?



Perspective

Received: 29 March 2018

Revised: 6 July 2018

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Gene drive systems: do they have a place in agricultural weed management?

Paul Neve*

SCI

Gene drives in plants: opportunities and challenges for weed control and engineered resilience

Luke G. Barrett^{1,3}, Mathieu Legros^{1,3}, Nagalingam Kumaran², Donna Glassop¹, S. Raghu² and Donald M. Gardiner¹

¹CSIRO Agriculture and Food, Canberra, Australian Capital Territory, Australia

Debate!

PRO vs ANTI chat wall

Meiotic drive to wipe out a weed that devastates tomato farms?
Pick a side. Post your best argument. React to the other side's posts.



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/debate-play.html?poll=pls152-tg2-drive-debate](https://greymonroe.github.io/live-poll-test/debate-play.html?poll=pls152-tg2-drive-debate)

[host screen](#)

Live poll

Multiple choice

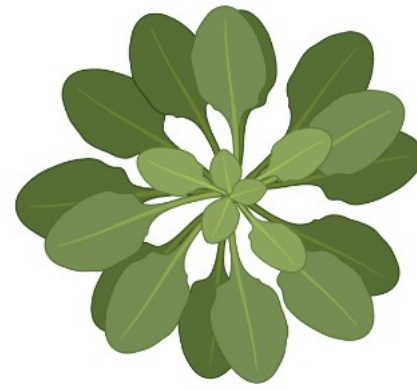
Ballot initiative: a weed causes major harm to tomato farmers, and scientists propose using meiotic drive to eliminate it. How do you vote?



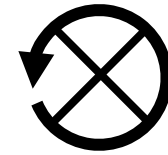
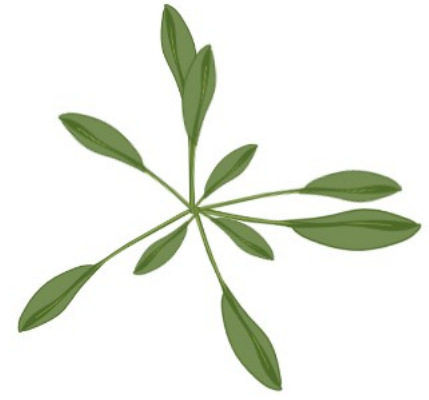
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[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-drive](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-drive)

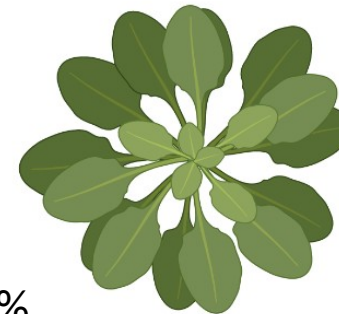
[host screen](#)



X



F1



F2

100%

Live poll

Text wall

Self the normal F1 and 100% of the F2 look normal. Where did the recessive phenotype go? Best idea wins.

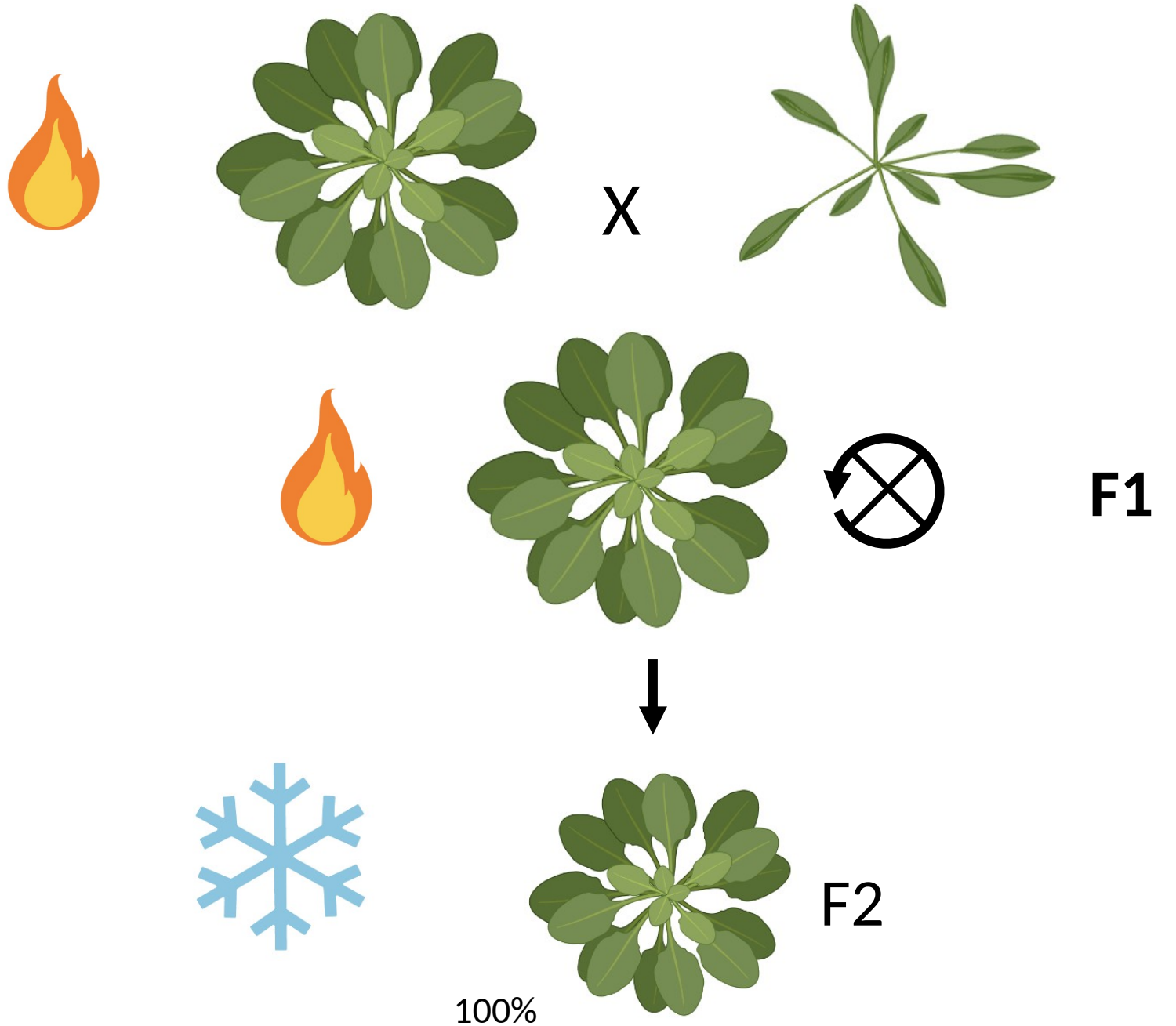


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[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-envf2](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-envf2)

[host screen](#)

Environmental dependance

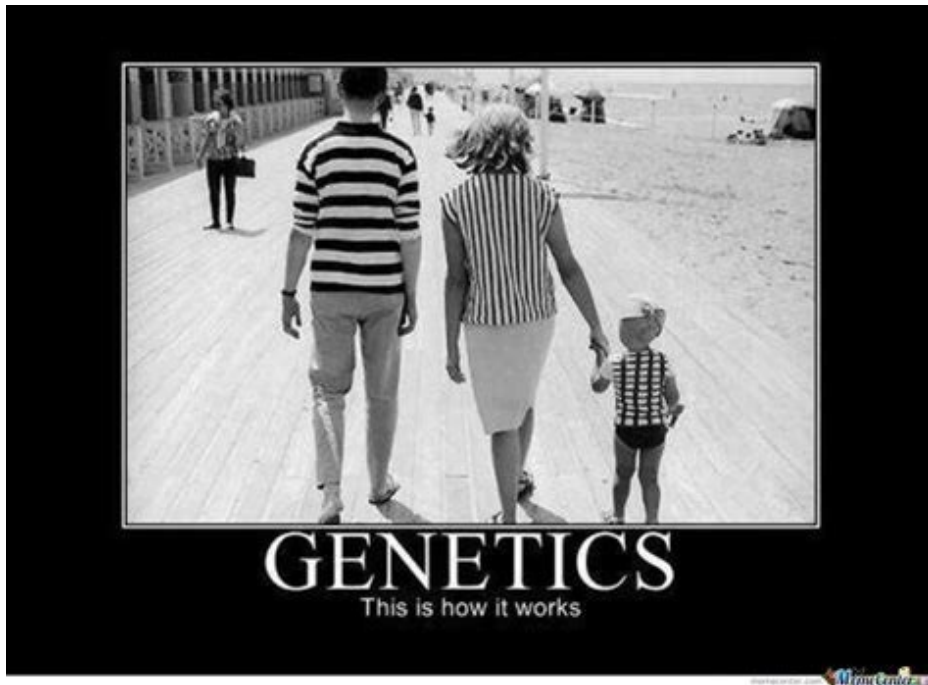


E.g. Heat-shock protein mutant

Live poll

Multiple choice

What kind of dominance is this?



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-stripes](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-stripes)

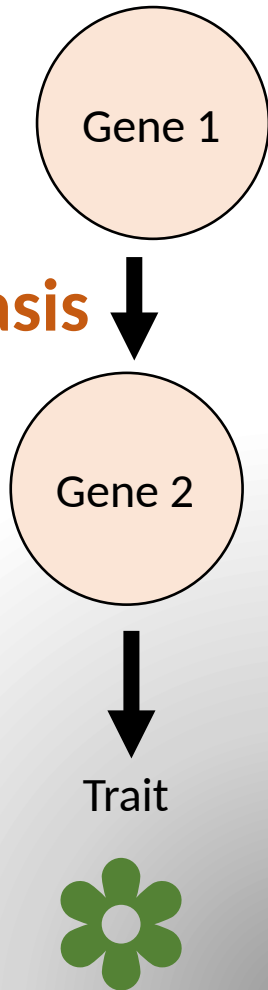
[host screen](#)

A photograph of a forest floor with tall, thin trees and sunlight filtering through the canopy. A large, semi-transparent white oval is centered over the image, containing the text "Break 😊".

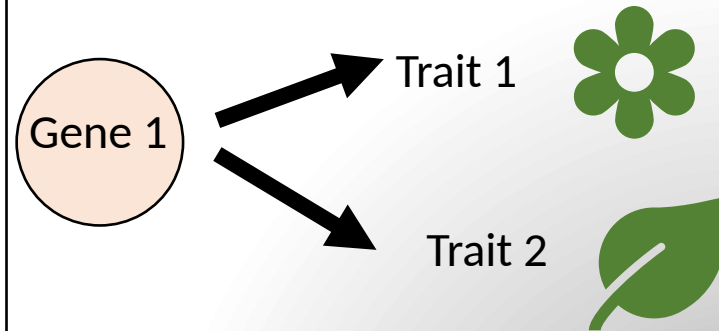
Break 😊

Non-independence between genes and traits

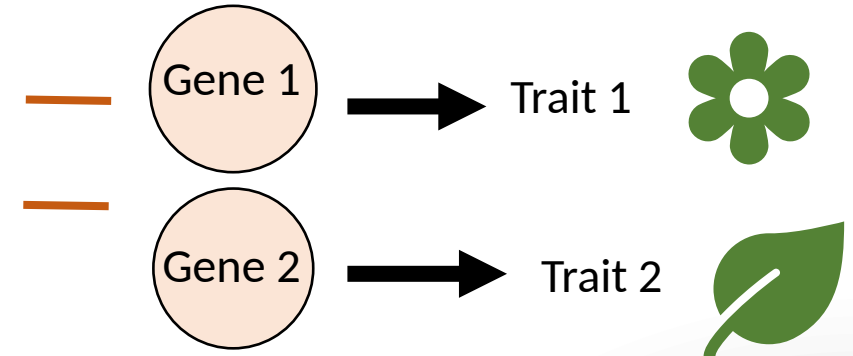
1. Epistasis



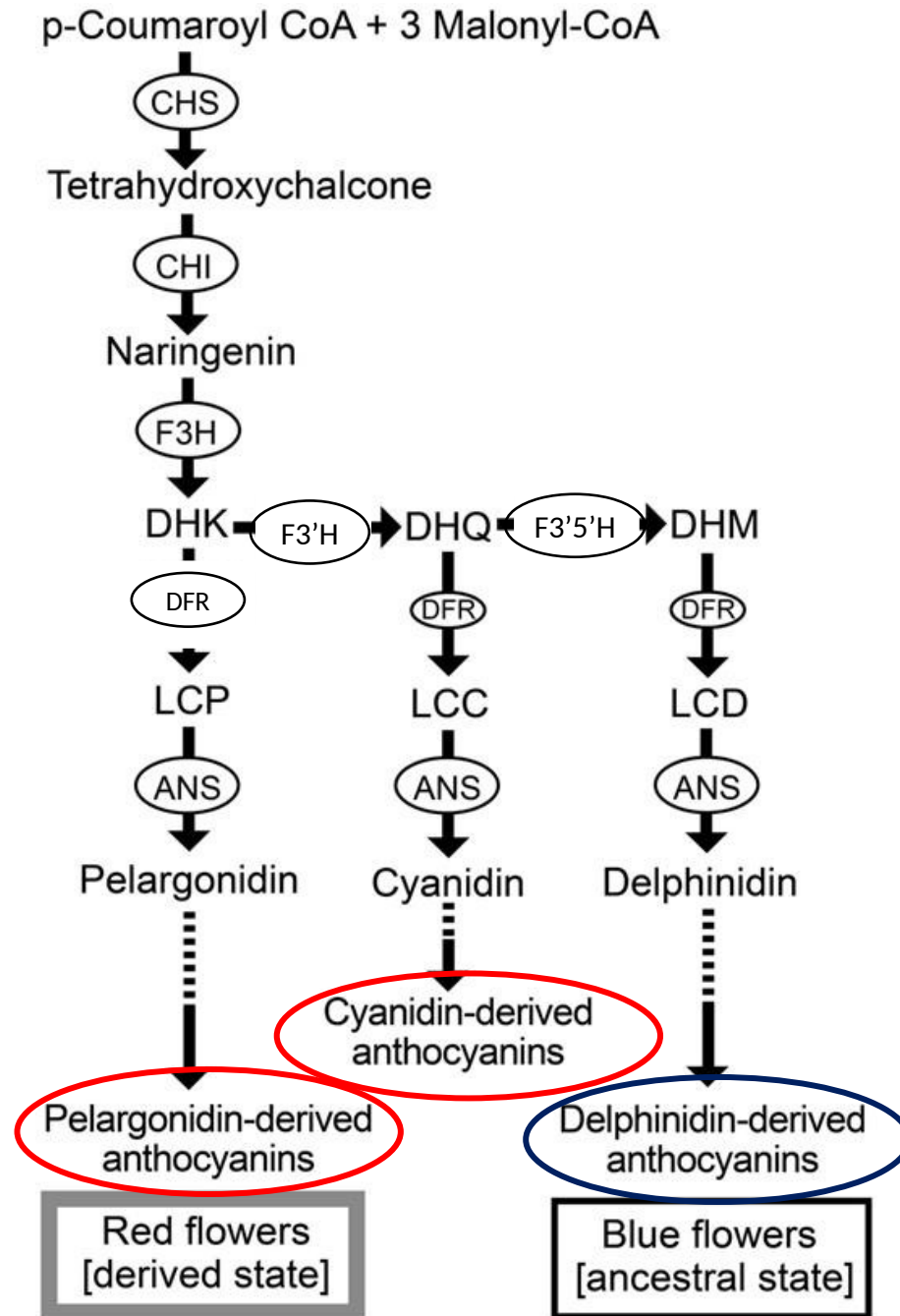
2. Pleiotropy



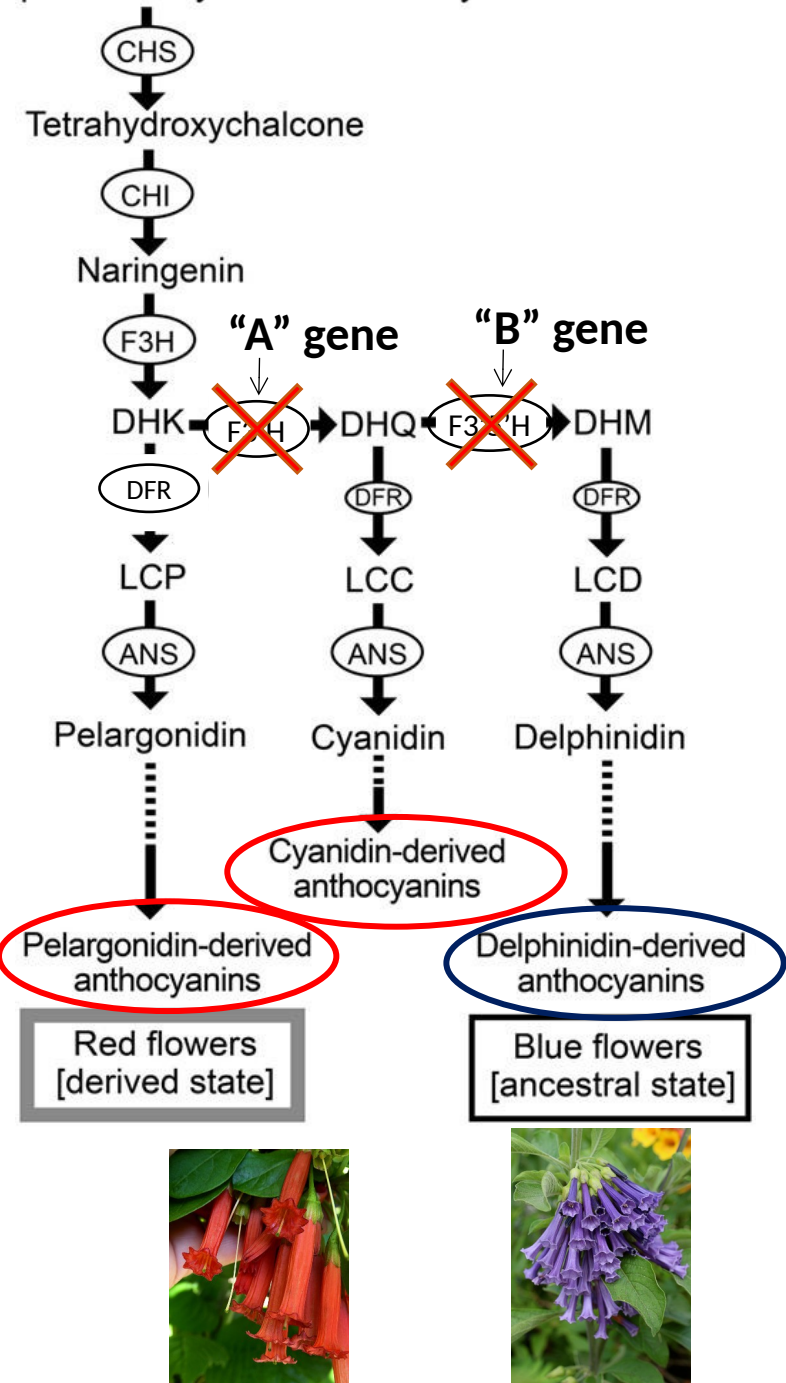
3. Linkage



Metabolic Network



p-Coumaroyl CoA + 3 Malonyl-CoA



Epistasis

Ipomea



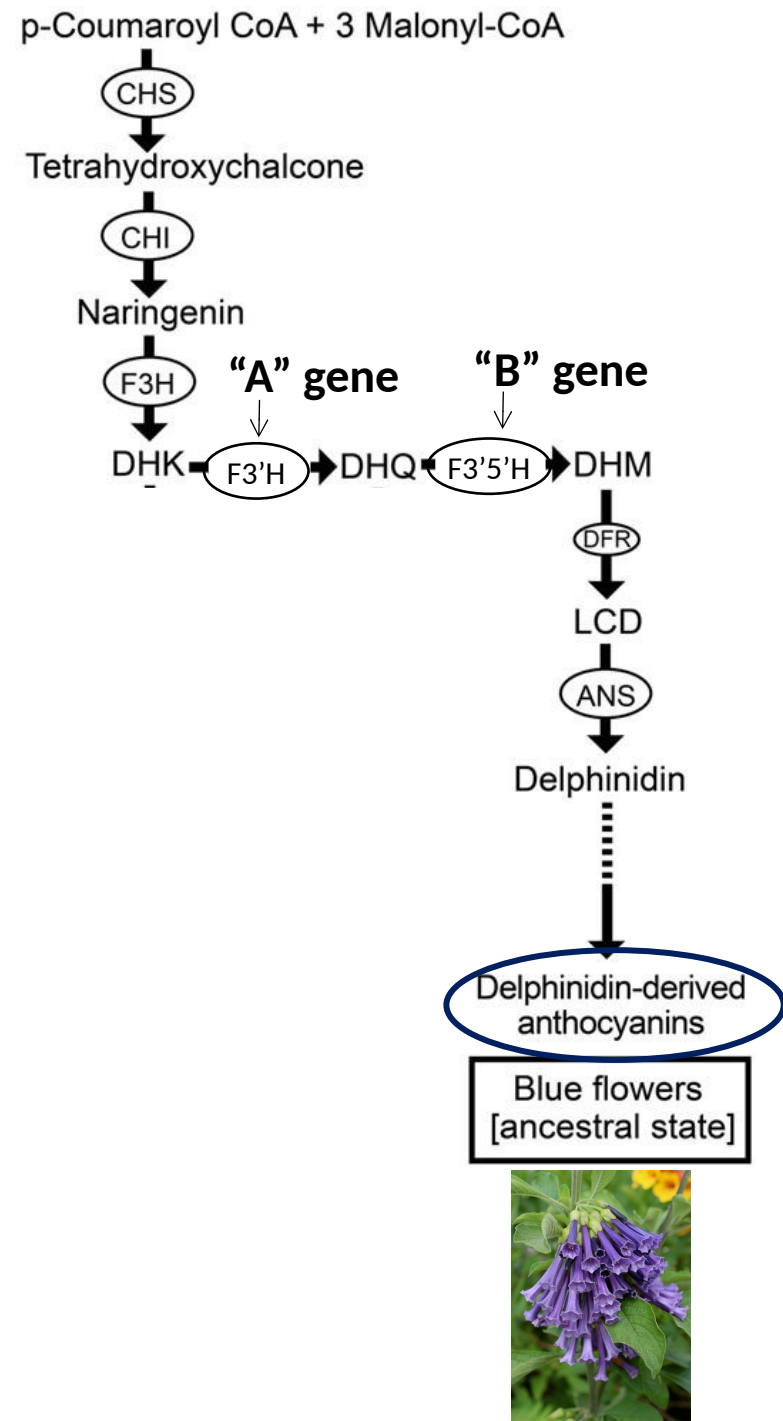
F1



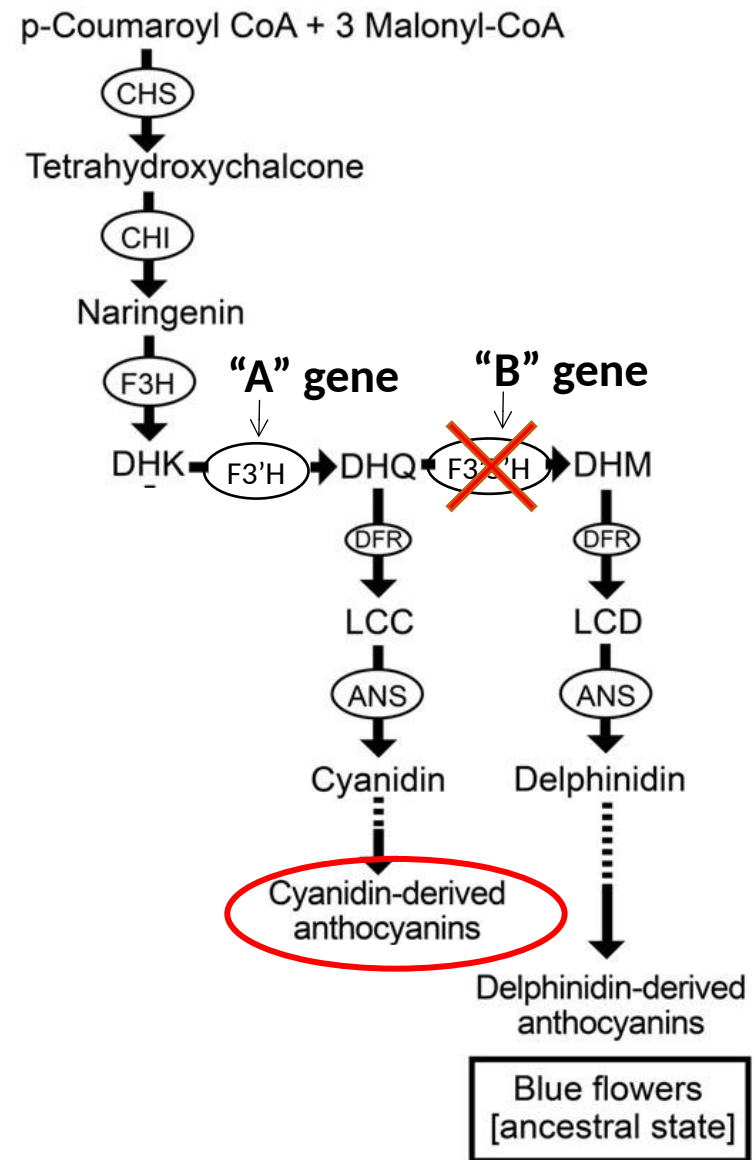
F2

9:7

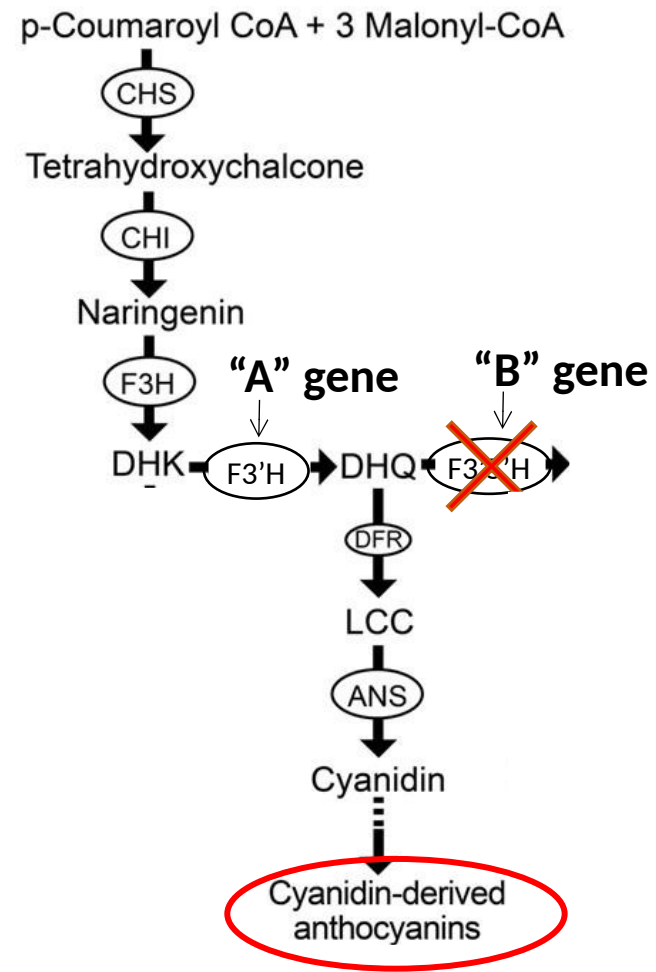
Epistasis



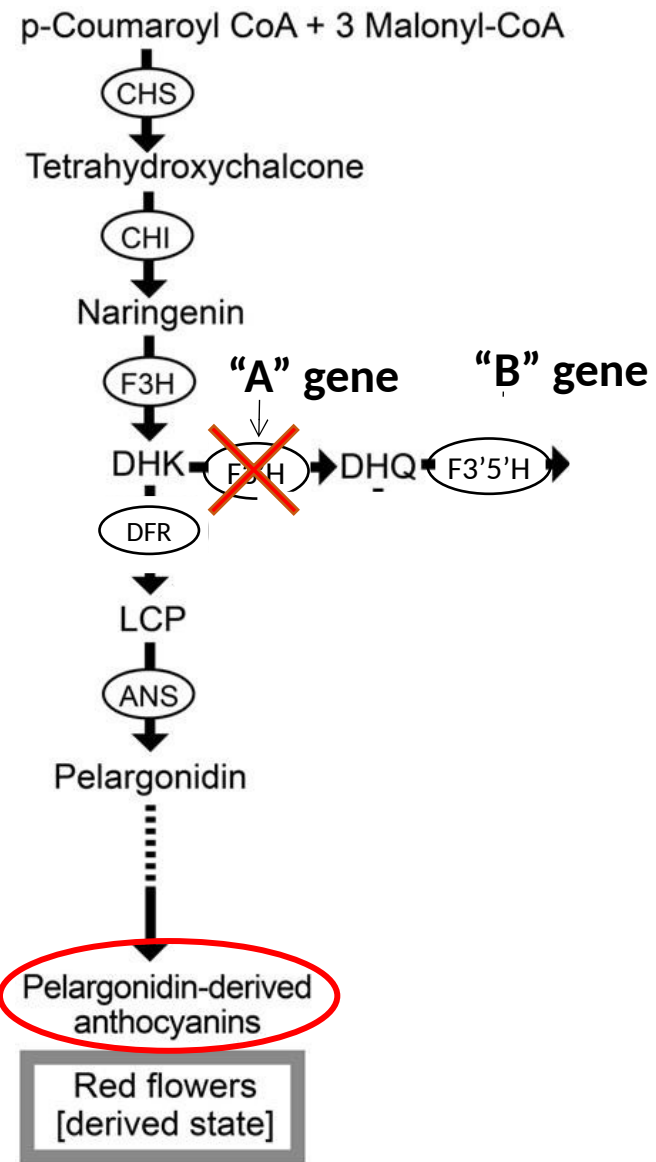
Epistasis



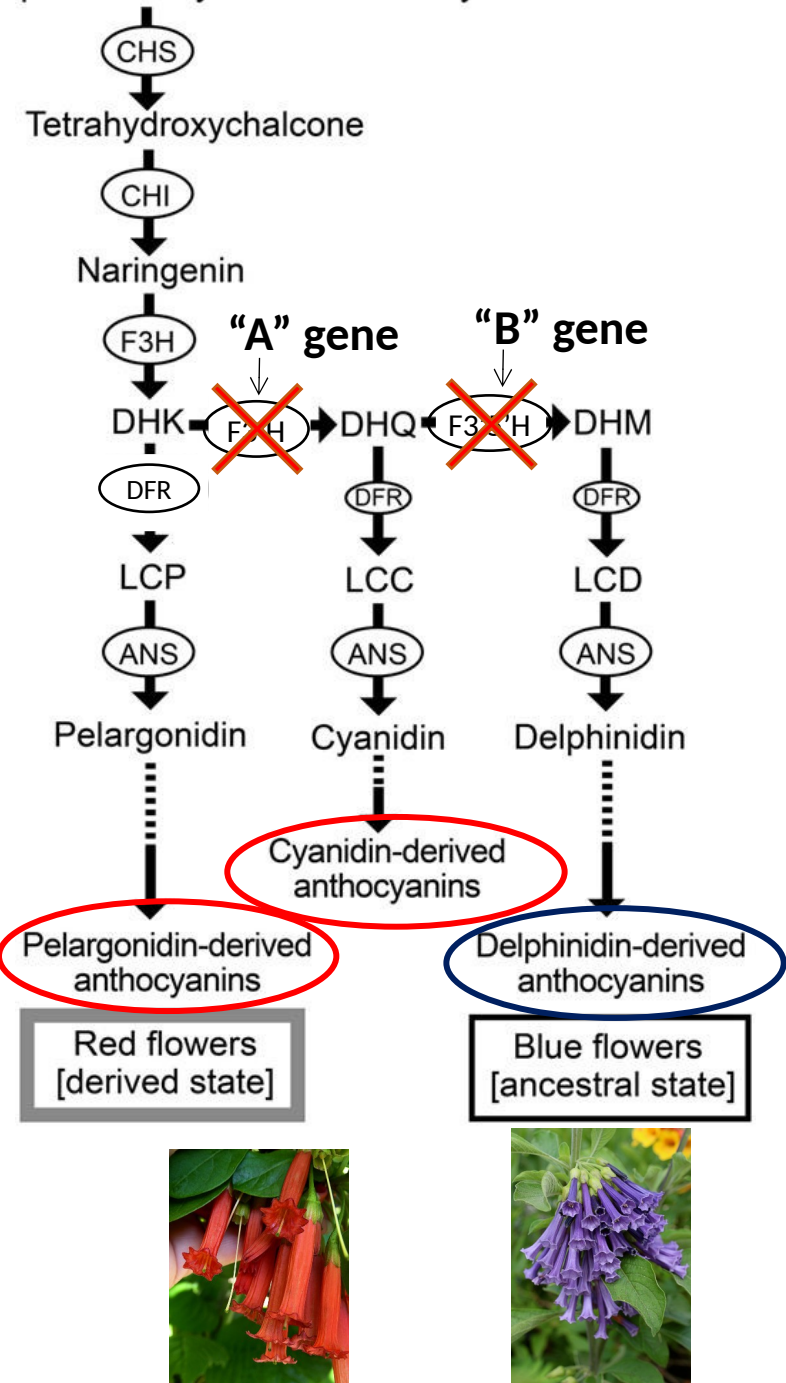
Epistasis



Epistasis



p-Coumaroyl CoA + 3 Malonyl-CoA



Epistasis

	A,B	A,b	a,B	a,b
A,B				
A,b				
a,B				
a,b				

AABB

AaBB

aaBB

AABb

AaBb

aaBb

AA**bb**

Aa**bb**

aabb

aa OR **bb** causes red phenotype

Ipomea



AABB

aabb



F1

AaBb

A-B-

aaB- or A-bb or aabb



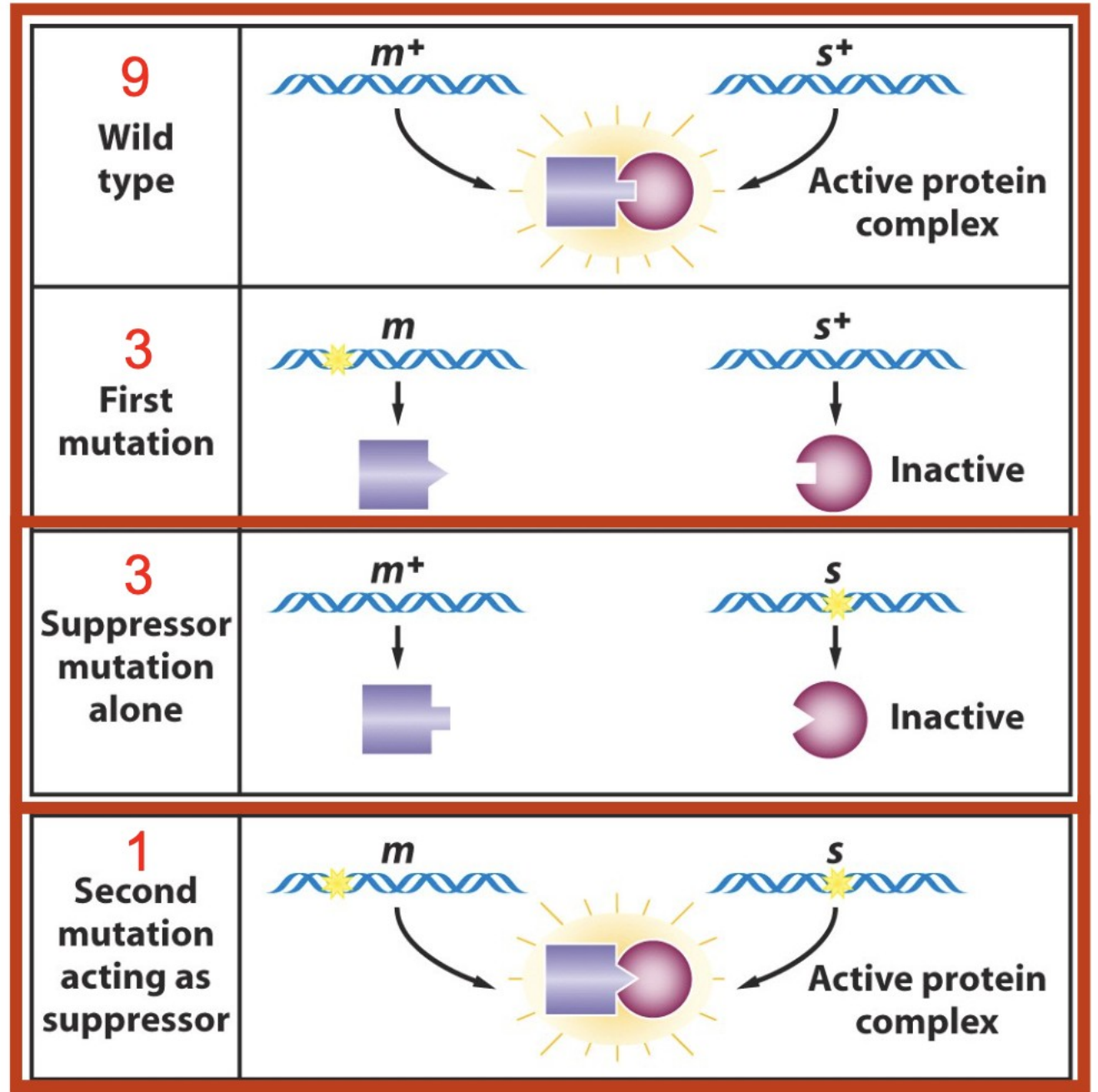
F2

9:7

Suppressor mutations F₂ ratio = 10:6



Green revolution wheat



Synthetic lethality: F2 ratio = 3:2 (9:6)

	A+ B+	A- B+	A+ B-	A- B-
A+ B+				
A- B+				
A+ B-				
A- B-				

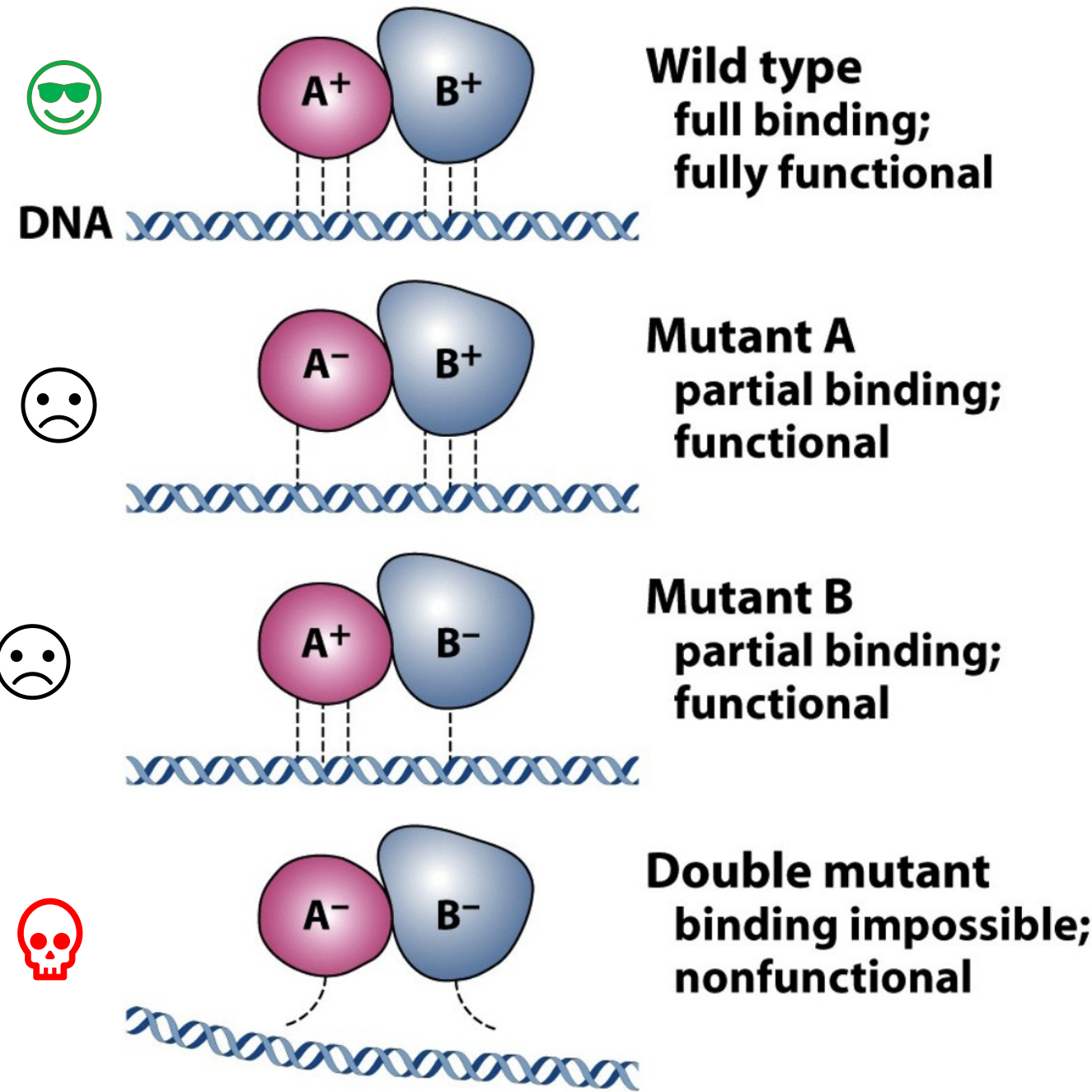


Figure 6-23
Introduction to Genetic Analysis, Ninth Edition
© 2008 W. H. Freeman and Company

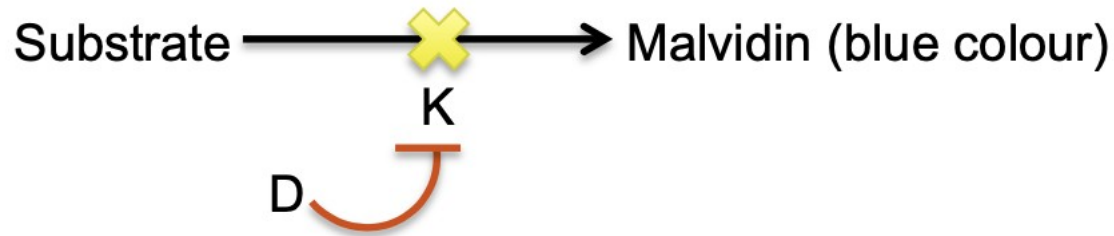


Primula

What is the expected fraction of blue offspring from a dihybrid cross:

$KkDd \times KkDd$

Hints: make dihybrid Punnet square



- Functional K allele needed to make blue pigment
- Functional D allele prevents K from acting

Live poll

Multiple choice

Primula, KkDd × KkDd: what fraction of the offspring are blue?



Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-primula](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-primula)

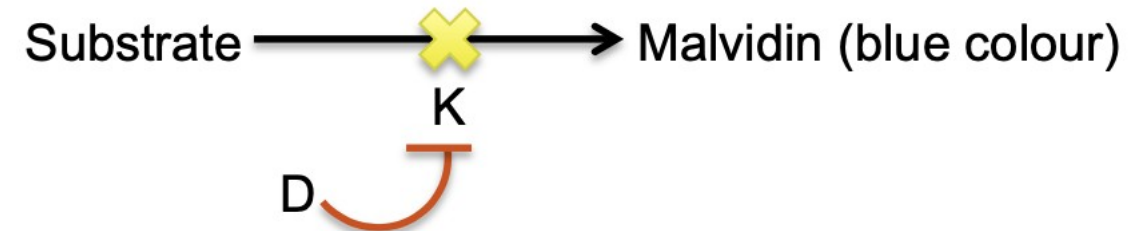
[host screen](#)

Dominant epistasis



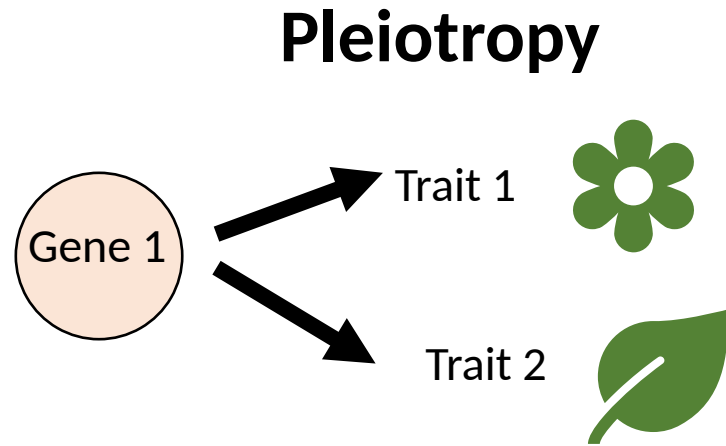
Primula

		Female Gametes			
		<i>KD</i>	<i>Kd</i>	<i>kD</i>	<i>kd</i>
Male Gametes	<i>KD</i>	<i>KKDD</i>	<i>KKDd</i>	<i>KkDD</i>	<i>KkDd</i>
	<i>Kd</i>	<i>KKDd</i>	<i>KKdd</i>	<i>KkDd</i>	<i>Kkdd</i>
	<i>kD</i>	<i>KkDD</i>	<i>KkDd</i>	<i>kkDD</i>	<i>kkDd</i>
	<i>kd</i>	<i>KkDd</i>	<i>Kkdd</i>	<i>kkDd</i>	<i>kkdd</i>

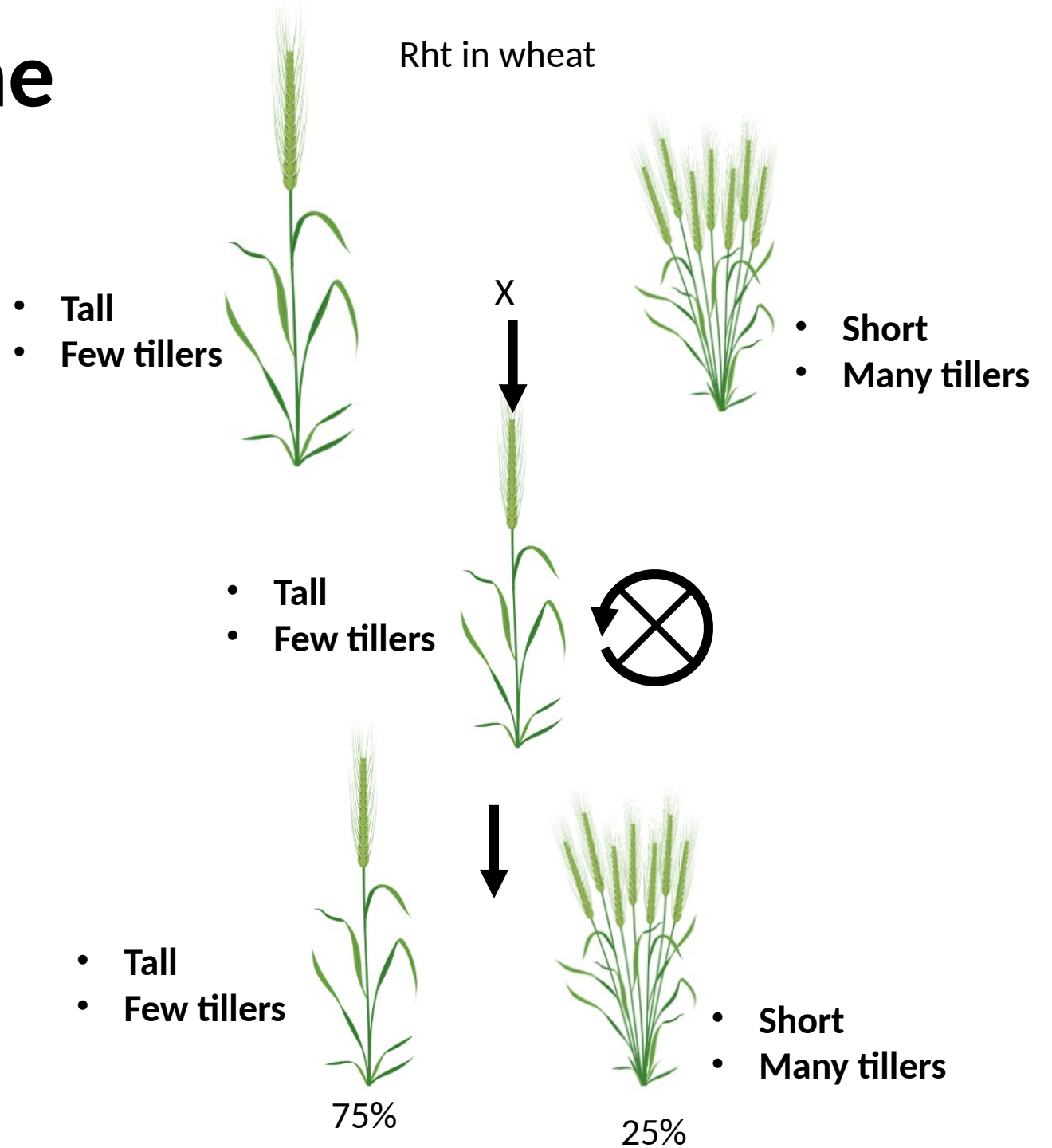
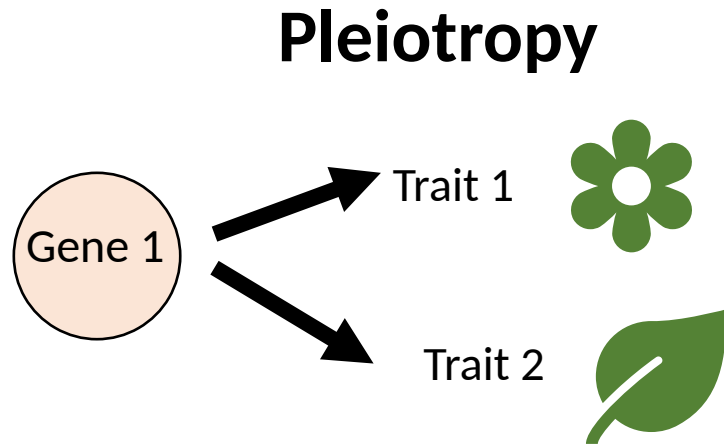


- Functional K allele needed to make blue pigment
- Functional D allele prevents K from acting

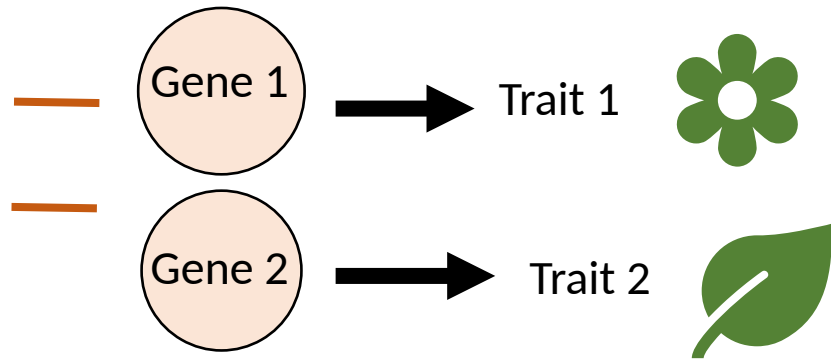
Pleiotropy: one gene multiple traits



Pleiotropy: one gene multiple traits



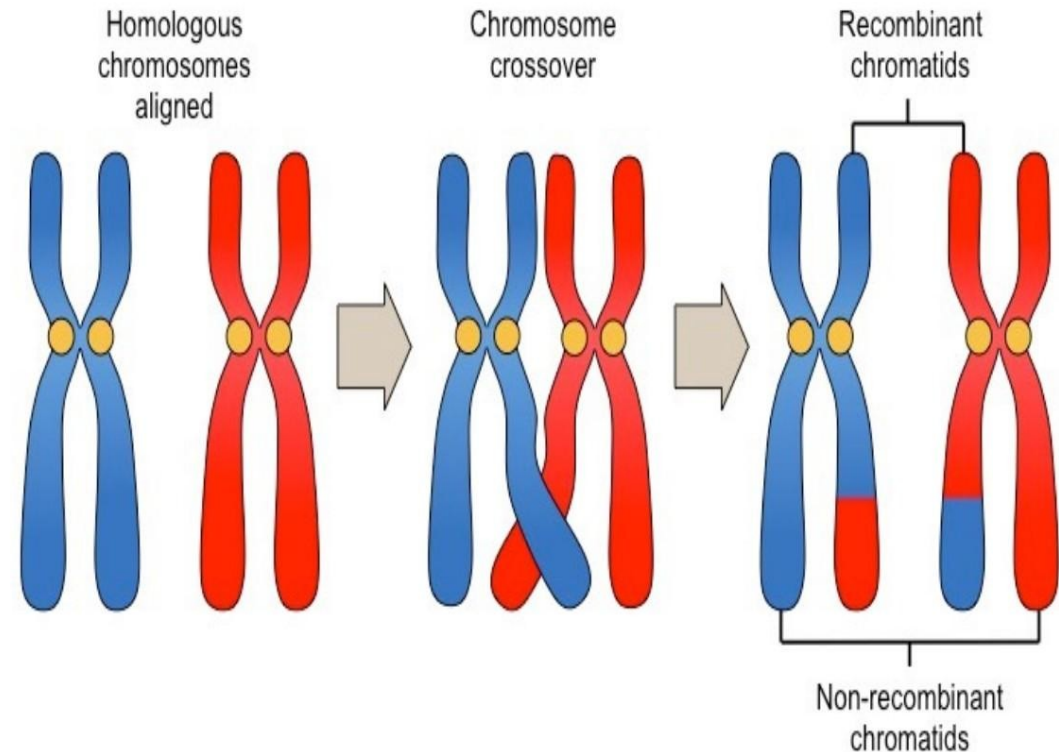
Linkage: two genes on the same chromosome

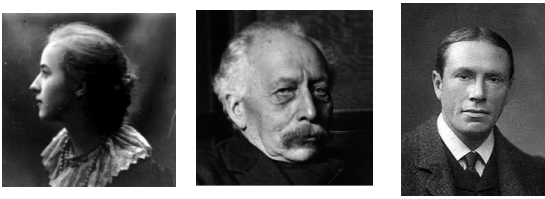


Linkage

Marvel: "Infinity War is the most ambitious crossover event in history"

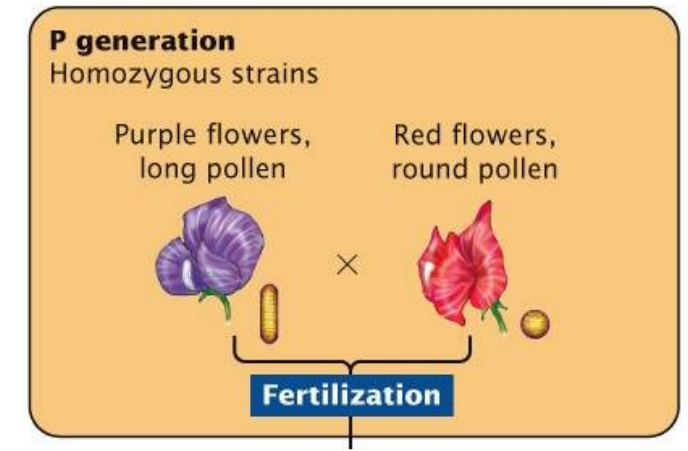
Me:

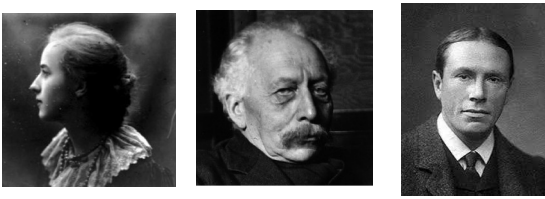




Edith Rebecca Saunders,
Reginald Punnet,
William Bateson

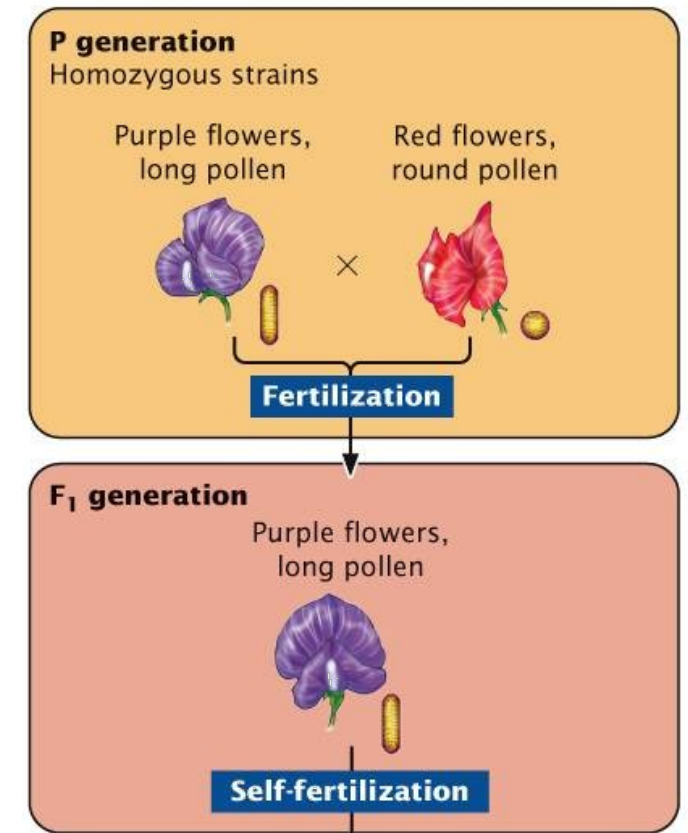
Cross two strains homozygous for two different traits.

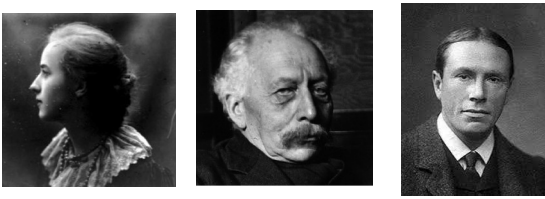




Edith Rebecca Saunders,
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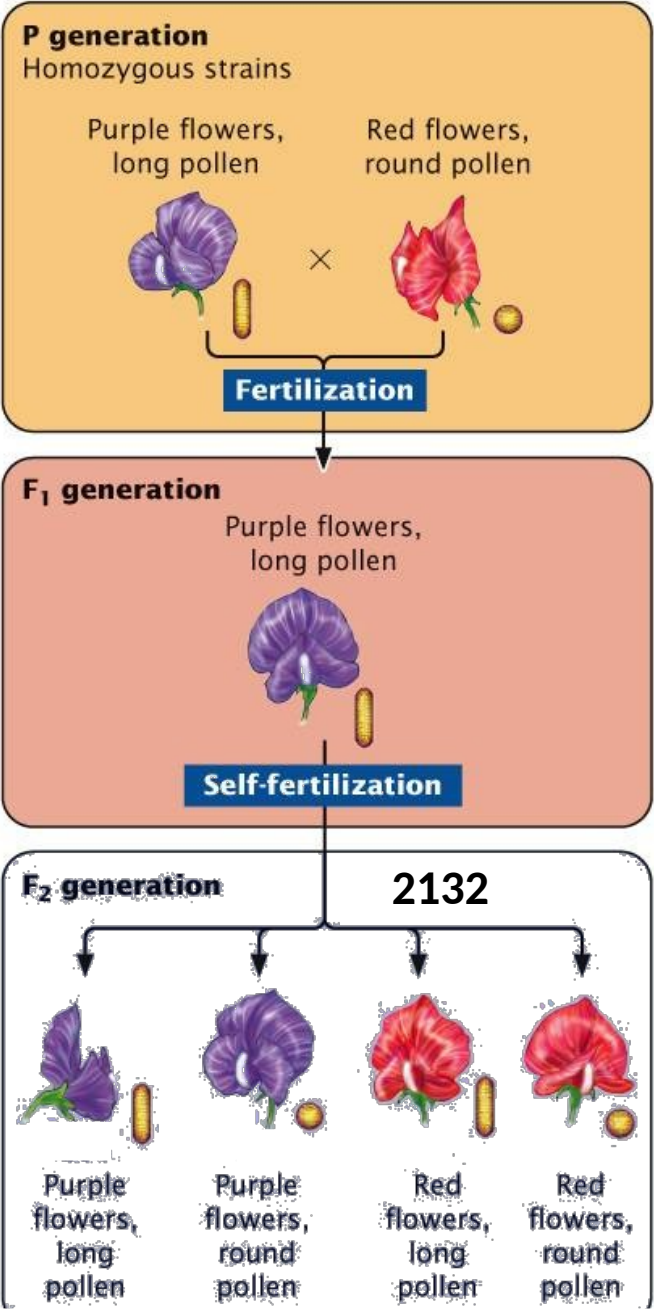


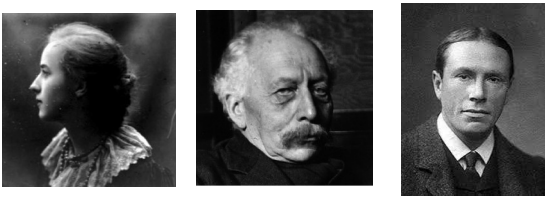


Edith Rebecca Saunders,
Reginald Punnet,
William Bateson

Phenotype	Expected
Purple, long	2132
Purple, round	
Red, long	
Red, round	
Total	

Cross two strains homozygous for two different traits.

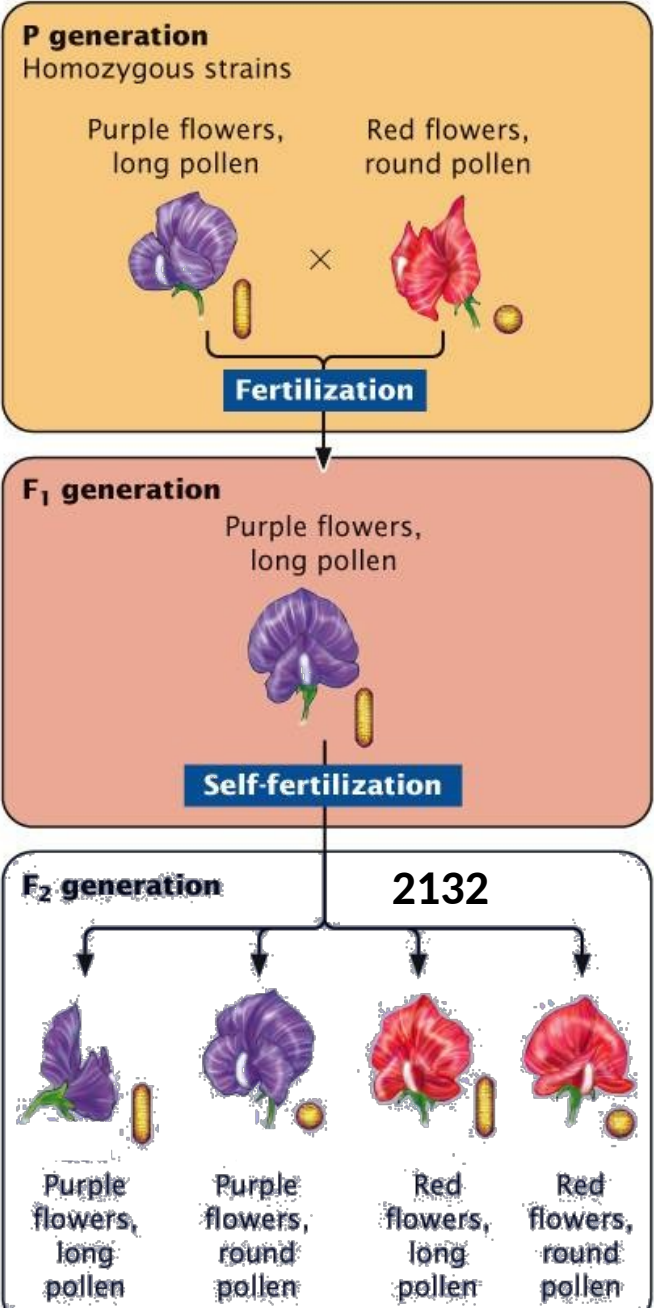


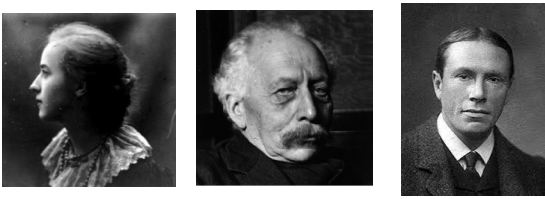


Edith Rebecca Saunders,
Reginald Punnett,
William Bateson

Phenotype	Expected
Purple, long	1199
Purple, round	400
Red, long	400
Red, round	133
Total	2132

Cross two strains homozygous for two different traits.



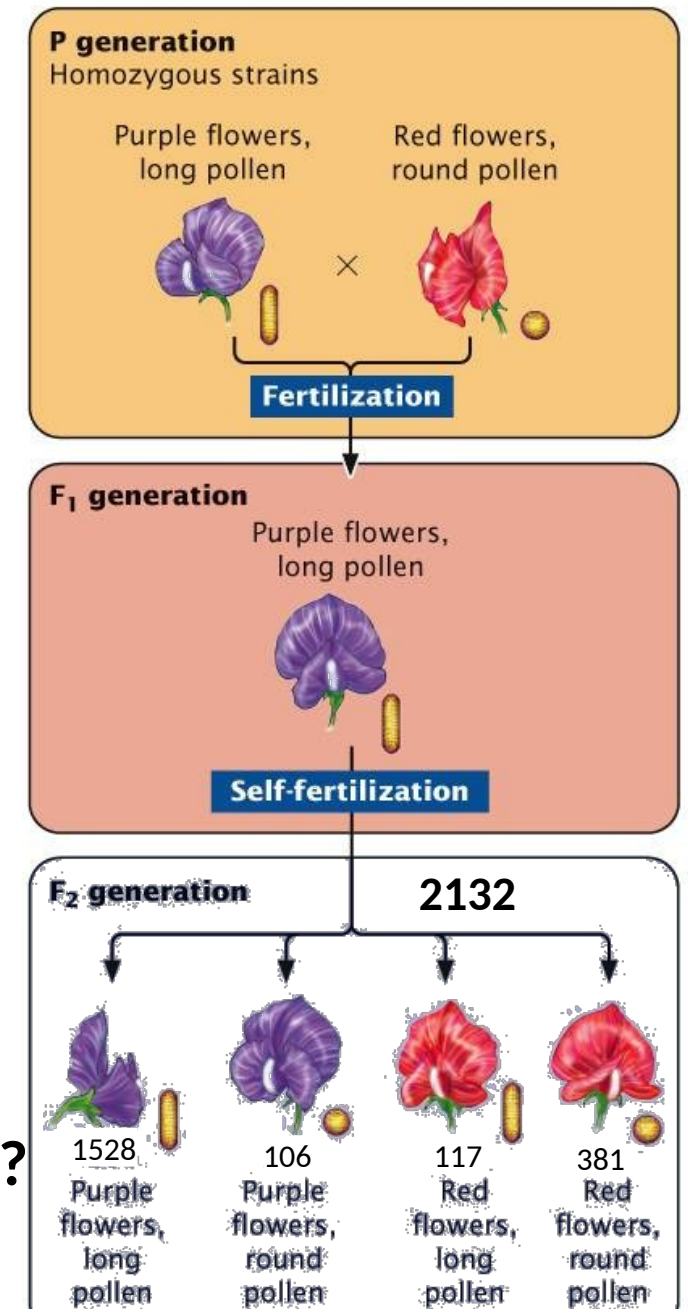


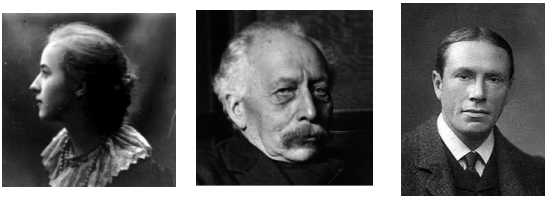
Edith Rebecca Saunders,
Reginald Punnett,
William Bateson

Phenotype	Expected	Observed
Purple, long	1199	1528
Purple, round	400	106
Red, long	400	117
Red, round	133	381
Total	2132	2132

Which phenotype combinations fewer observed than expected?
Which phenotype combinations are "recombinants"?

Cross two strains homozygous for two different traits.



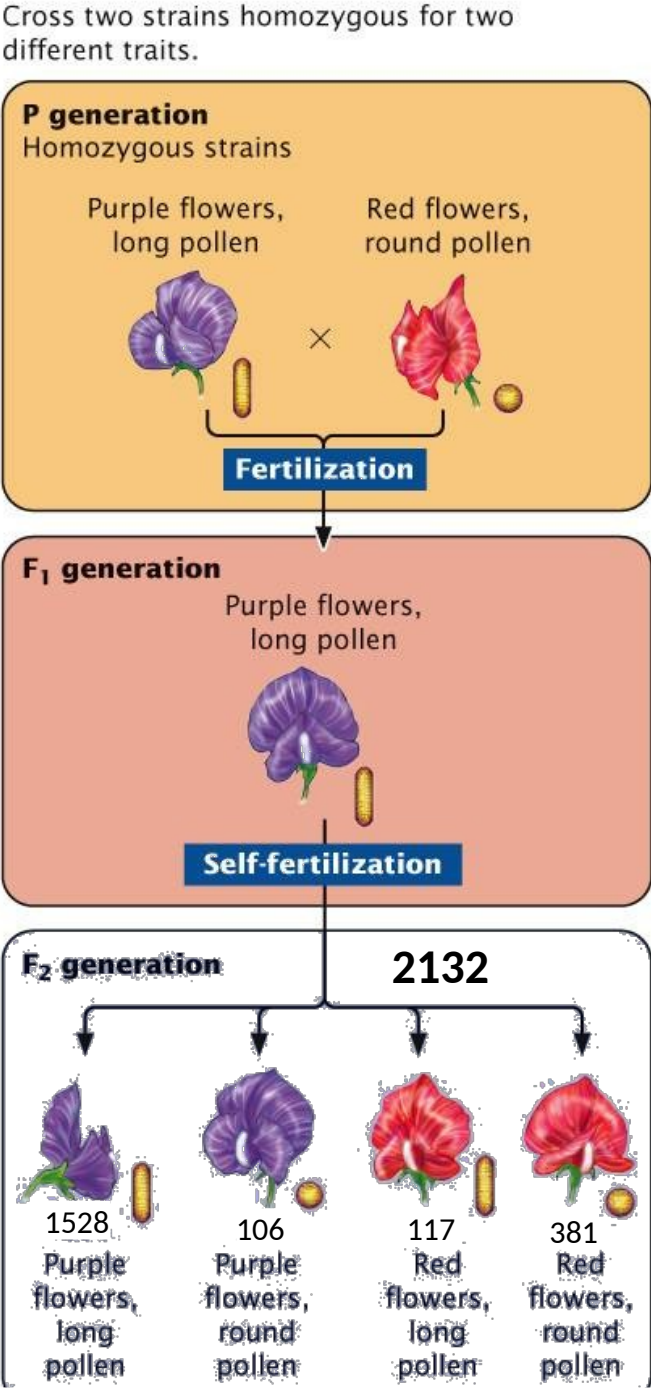


Edith Rebecca Saunders,
Reginald Punnett,
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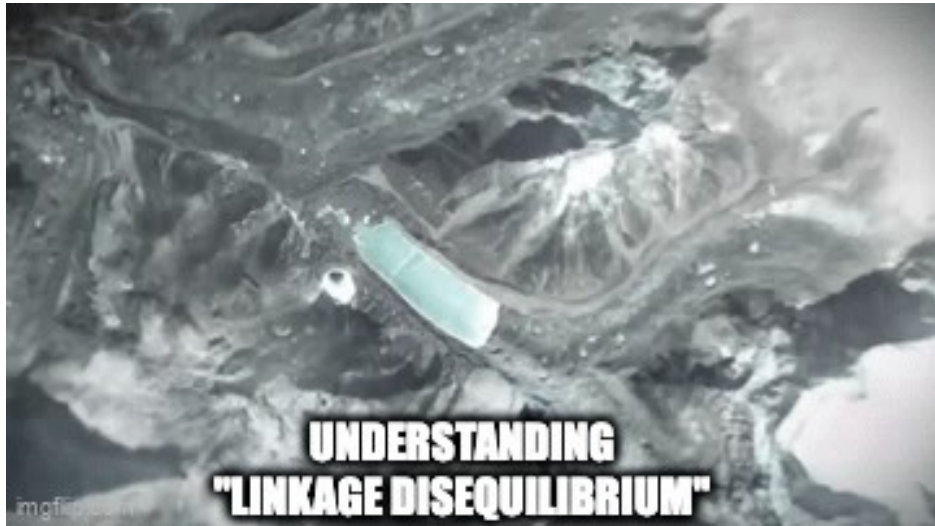
Alleles between genes “mix”
as expected... they are in....
"Equilibrium"

Phenotype	Expected	Observed
Purple, long	1199	1528
Purple, round	400	106
Red, long	400	117
Red, round	133	381
Total	2132	2132

Alleles between genes do not “mix”
as expected... they are in....
Linkage "DISEquilibrium"

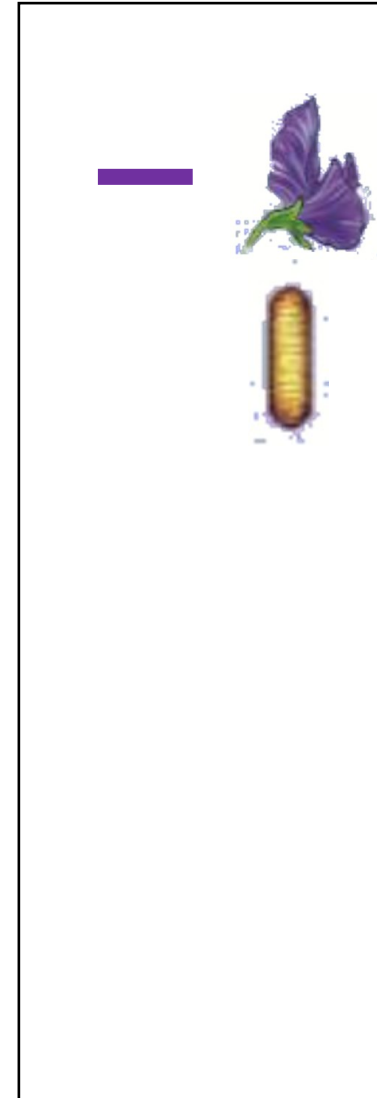
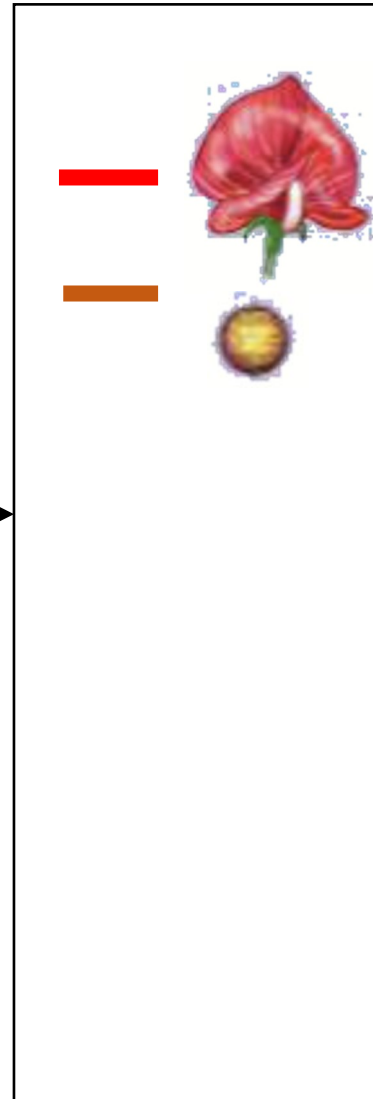
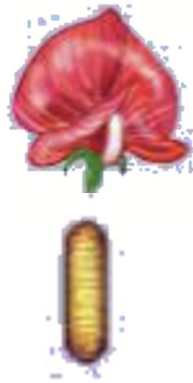
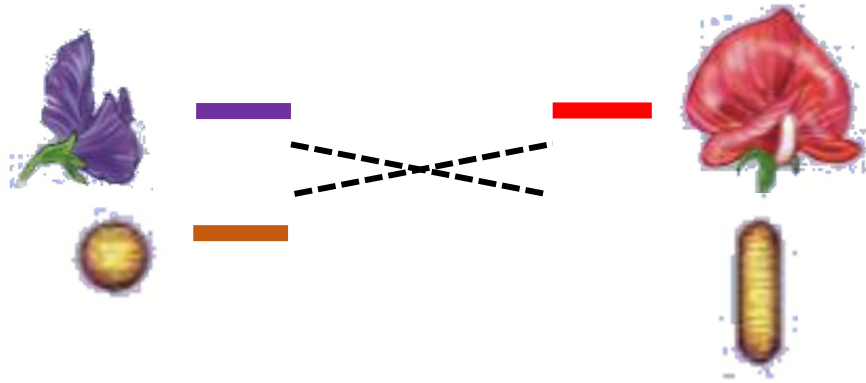


Genes close to each other on the same chromosome show **greater linkage disequilibrium (LD)** and **less recombination**



Meiosis in the F2

Gametes passed to offspring

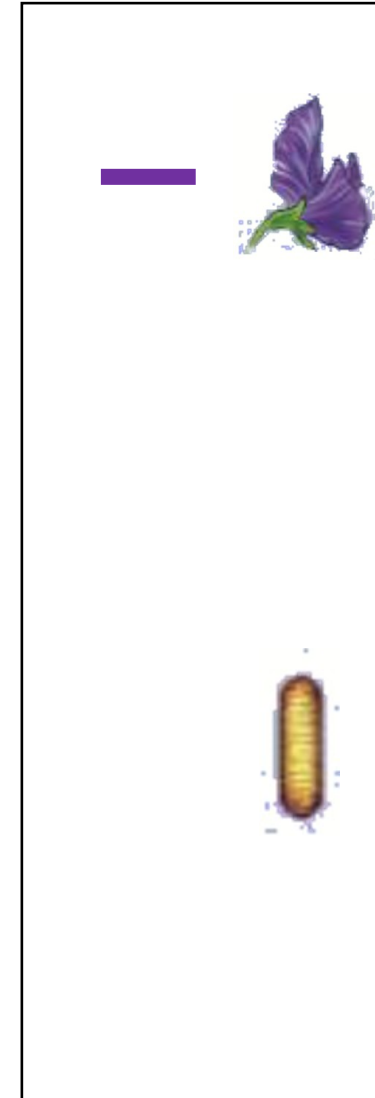
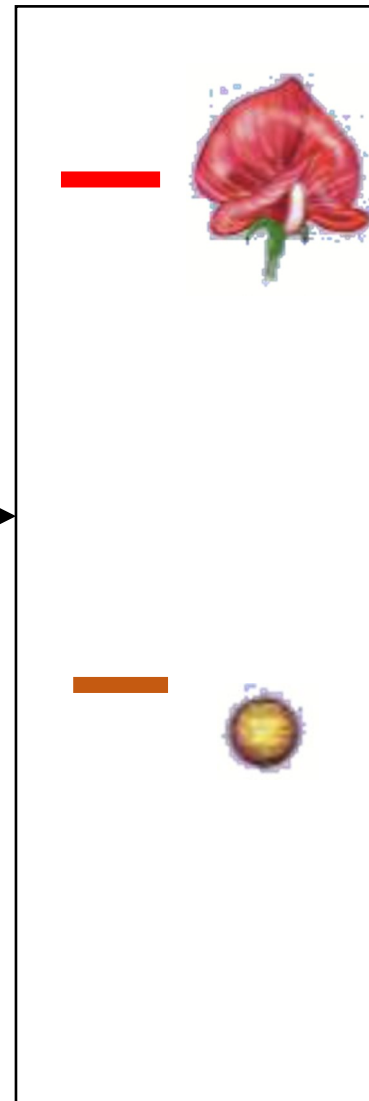
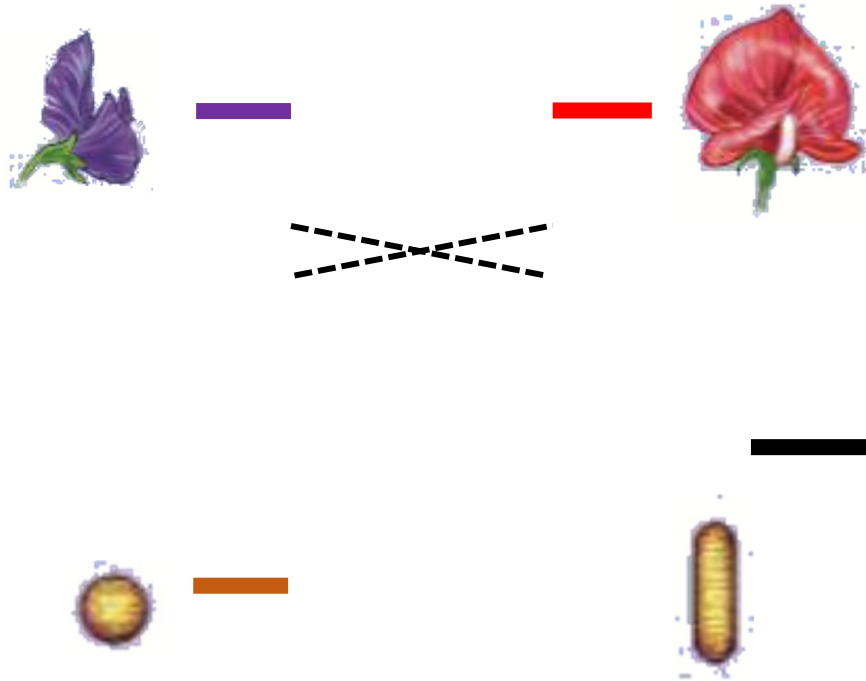


Recombinants:
Combinations
of alleles not
found in the
parental
generation

Recombination

Meiosis in the F2

Gametes passed to offspring



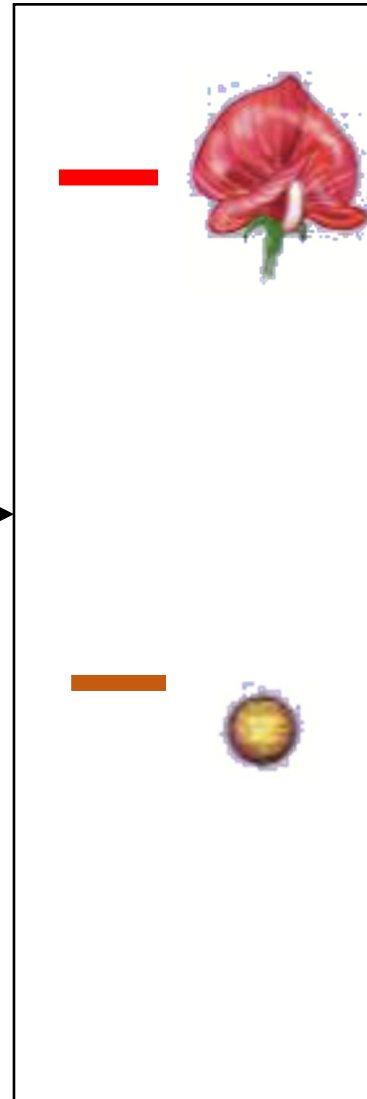
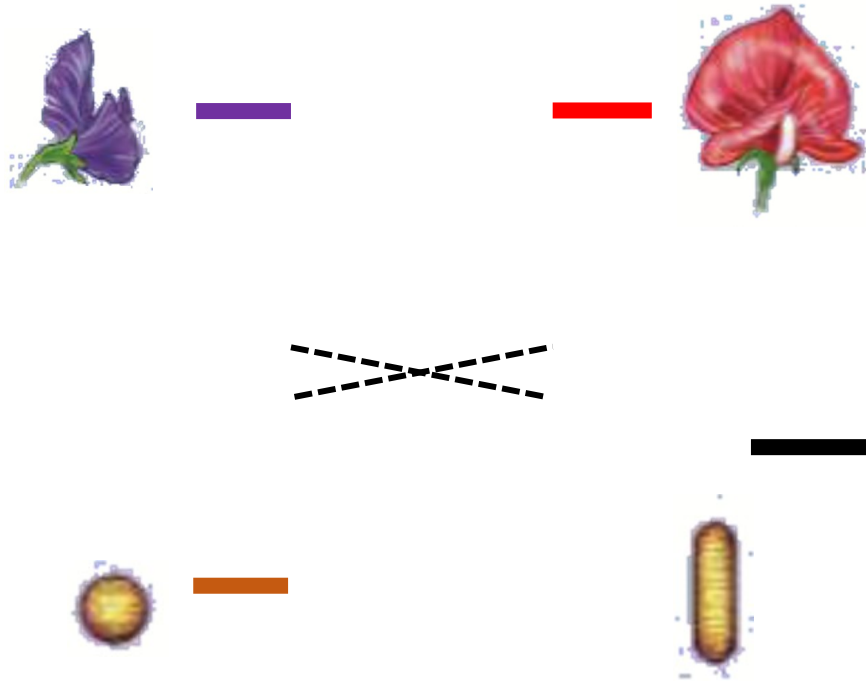
Recombinants:
Combinations
of alleles not
found in the
parental
generation

Further the genes, the more opportunities for

Recombination

Meiosis in the F2

Gametes passed to offspring



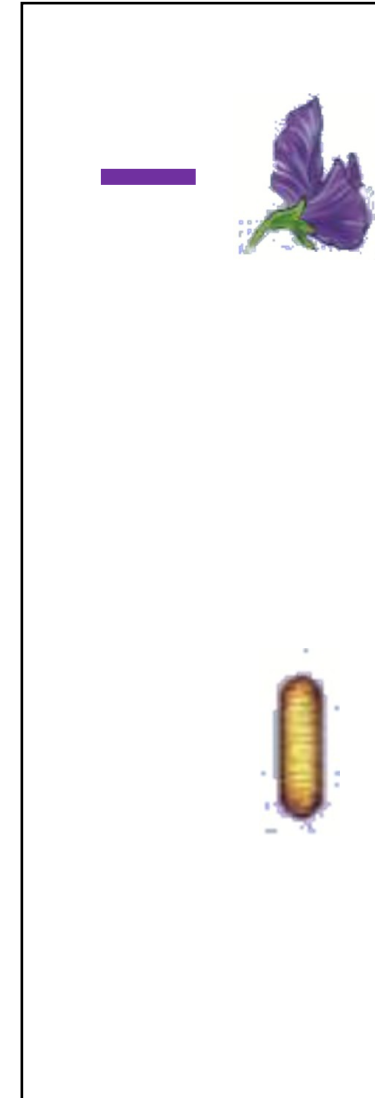
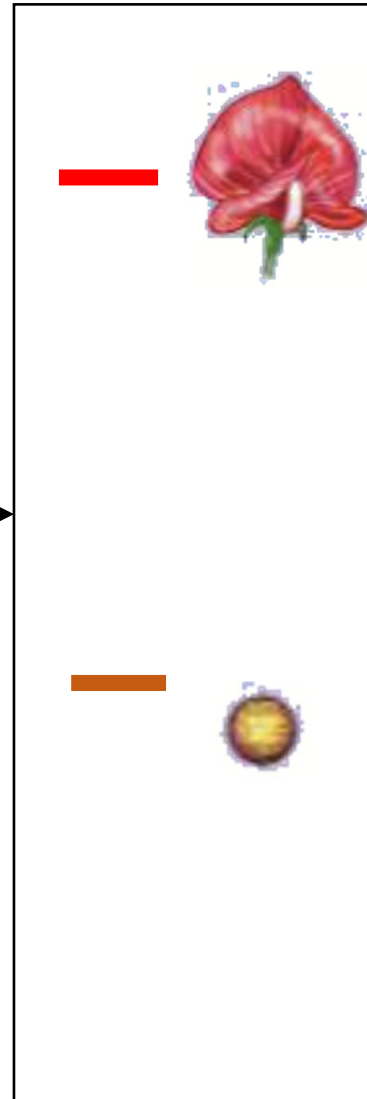
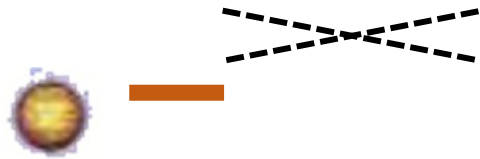
Recombinants:
Combinations
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Further the genes, the more opportunities for

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Meiosis in the F2

Gametes passed to offspring



Recombinants:
Combinations
of alleles not
found in the
parental
generation

Further the genes, the more opportunities for

Recombination

Meiosis in the F2

Gametes passed to offspring

A — — T

C — T



— T

— C

— A

T

Recombinants:
Combinations
of alleles not
found in the
parental
generation

Further the genes, the more opportunities for

Recombination

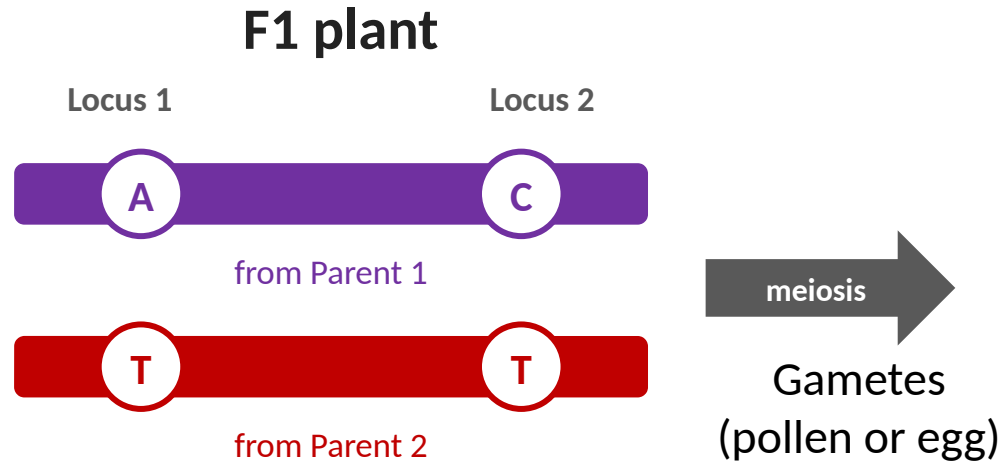
Recombination: seeing it in DNA

Where it happens • how we count it • how it builds a map

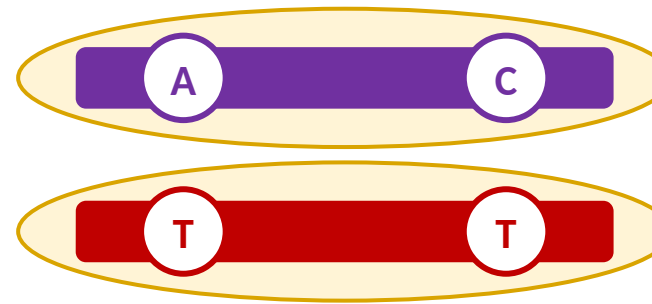


Crossovers happen during meiosis in the F1

The F1 carries one chromosome from each parent

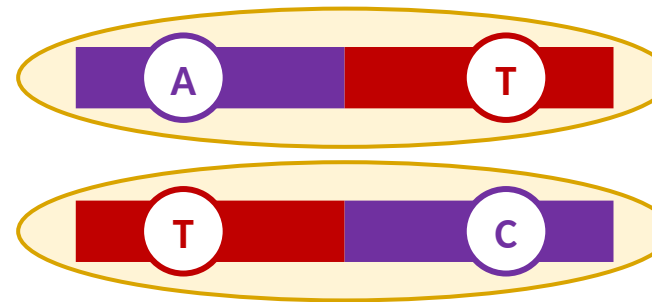


No crossover between the loci



Parental gametes (pollen)
same combinations as the
parents

A crossover between the loci

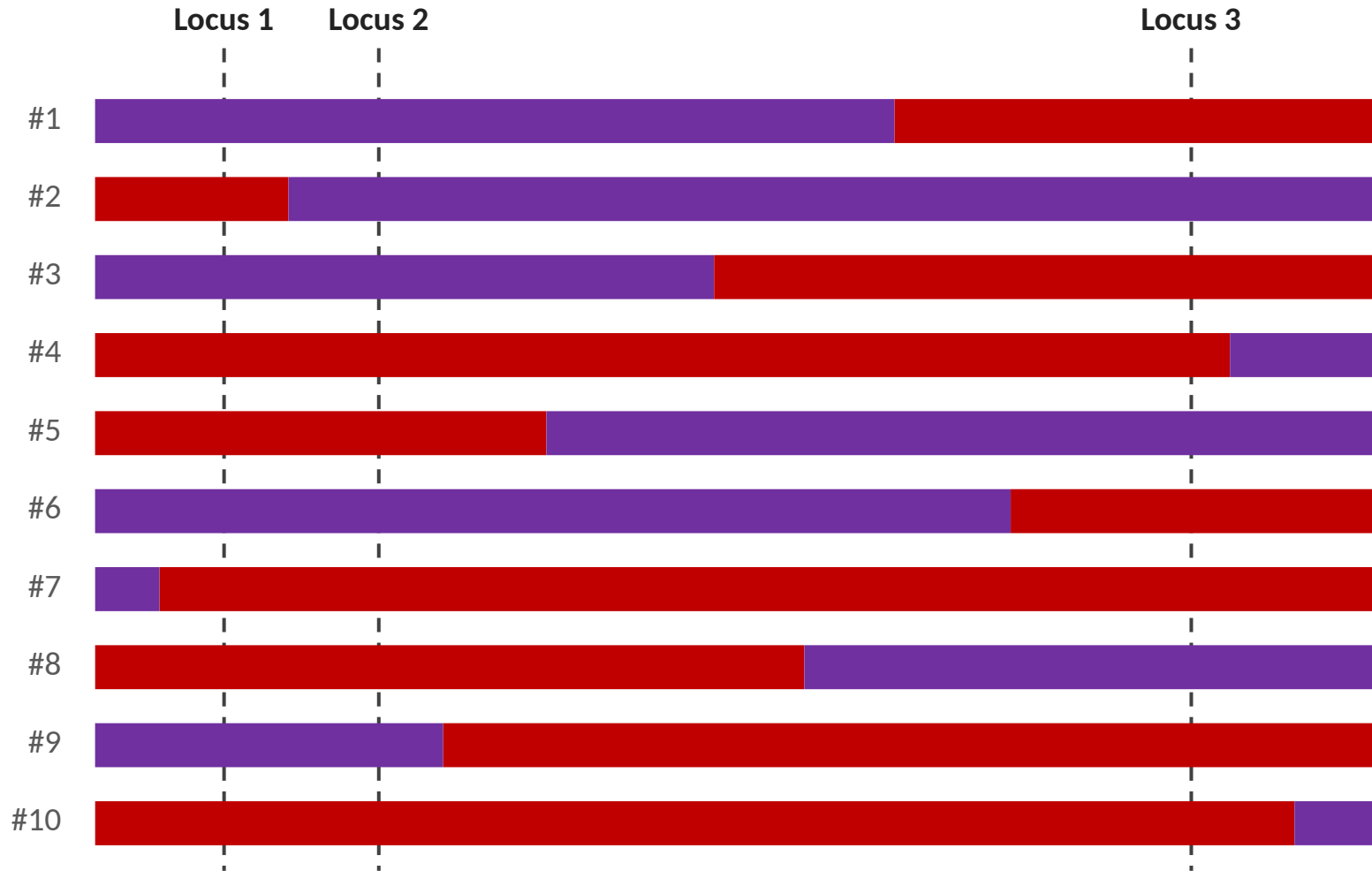


Recombinant gametes (pollen)
new combinations neither
parent had

Haplotype: the set of alleles on one chromosome.
Recombination makes new haplotypes.

Loci far apart split up more often

10 meioses in the F1, one crossover each, at a random spot

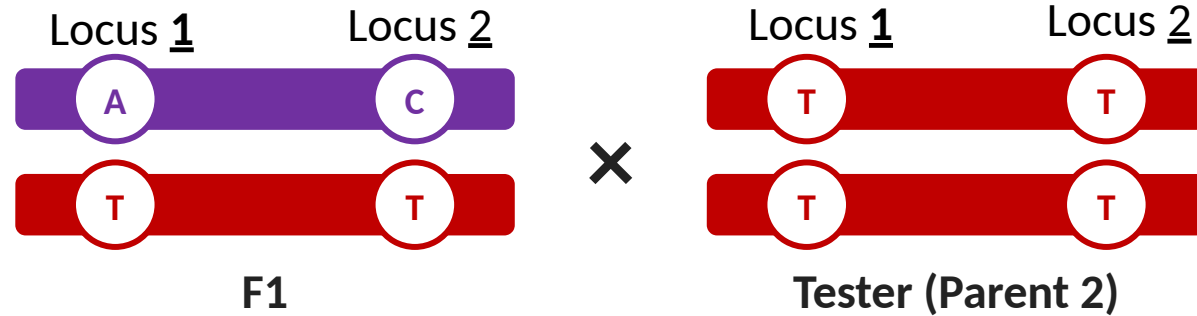


Loci 1 & 2 (close)
split apart in **1 of 10**

Loci 1 & 3 (far)
split apart in **7 of 10**

The trick: a testcross lets us read the F1's gametes

We can't watch crossovers, but we can genotype SNPs in the offspring



The tester can only give T T.
So each offspring's genotype tells us which gamete the F1 made.

Gamete from F1	+ from tester	Offspring SNPs Locus 1 Locus 2		Type
		AT	CT	Parental
		TT	TT	Parental
		AT	TT	Recombinant
		TT	CT	Recombinant

Your data: 20 testcross offspring

Each card = one offspring's SNP genotypes at Locus 1 and Locus 2. Which ones carry a recombinant chromosome from the F1?

#1 AT CT	#2 TT TT	#3 AT CT	#4 AT TT	#5 TT TT
#6 TT TT	#7 AT CT	#8 AT CT	#9 TT CT	#10 TT TT
#11 AT CT	#12 TT TT	#13 TT CT	#14 AT CT	#15 TT TT
#16 AT CT	#17 TT TT	#18 AT TT	#19 TT TT	#20 AT CT

Parental = **AT CT** or **TT TT** Recombinant = **AT TT** or **TT CT**

Live poll

Number: live histogram

How many of the 20 testcross offspring are recombinants?



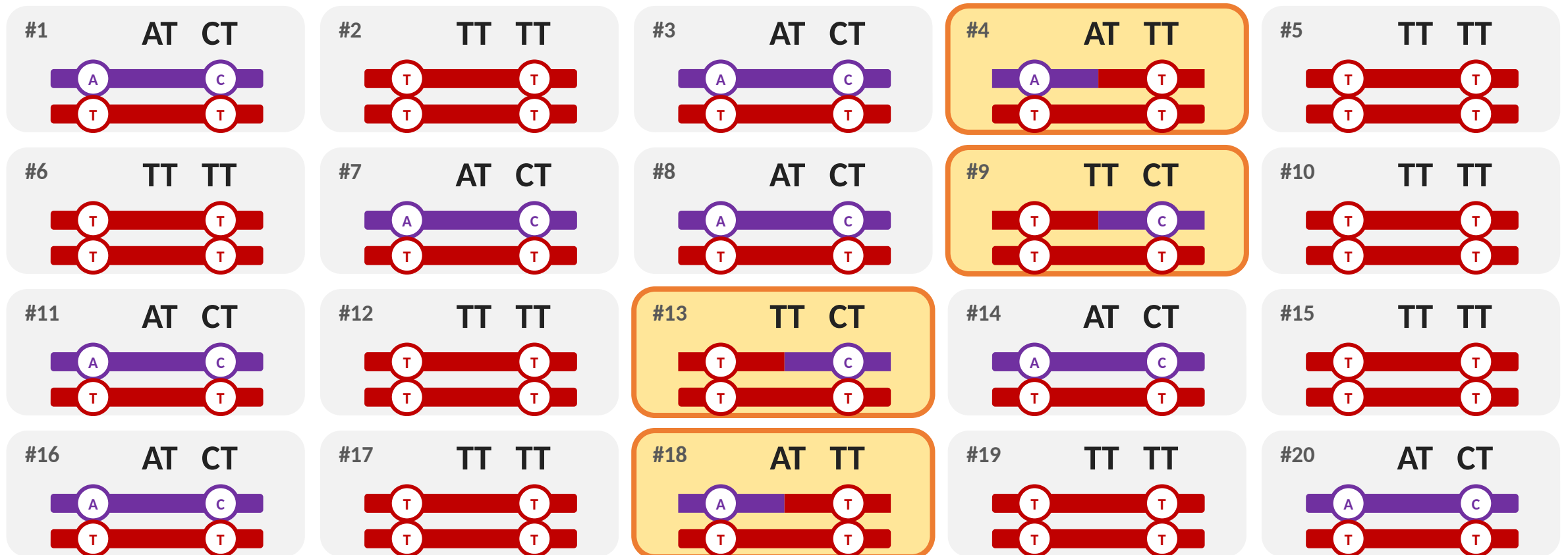
Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-rfcount](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-rfcount)

[host screen](#)

4 of 20 are recombinants

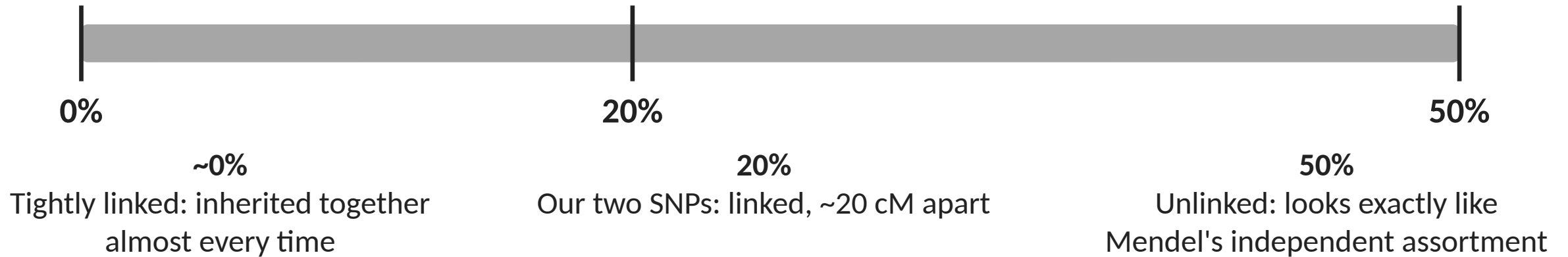
Each offspring is diploid: top chromosome from the F1, bottom (T T) from the tester



Recombination frequency = $4 / 20 = 20\%$ → ~20 cM apart

Recombination frequency is a distance

Recombination frequency (RF) = recombinant offspring $\div$ total offspring
1% recombination = 1 **centiMorgan (cM)** of genetic distance



Four markers: what order are they in?

Recombination frequency (%) between each pair of markers

	A	B	C	D
A	—	20%	10%	15%
B	20%	—	30%	5%
C	10%	30%	—	25%
D	15%	5%	25%	—

Hint: the biggest number tells you which two markers are on the ends.

Live poll

Multiple choice

Using the table of recombination frequencies, what is the order of markers A, B, C, D along the chromosome?



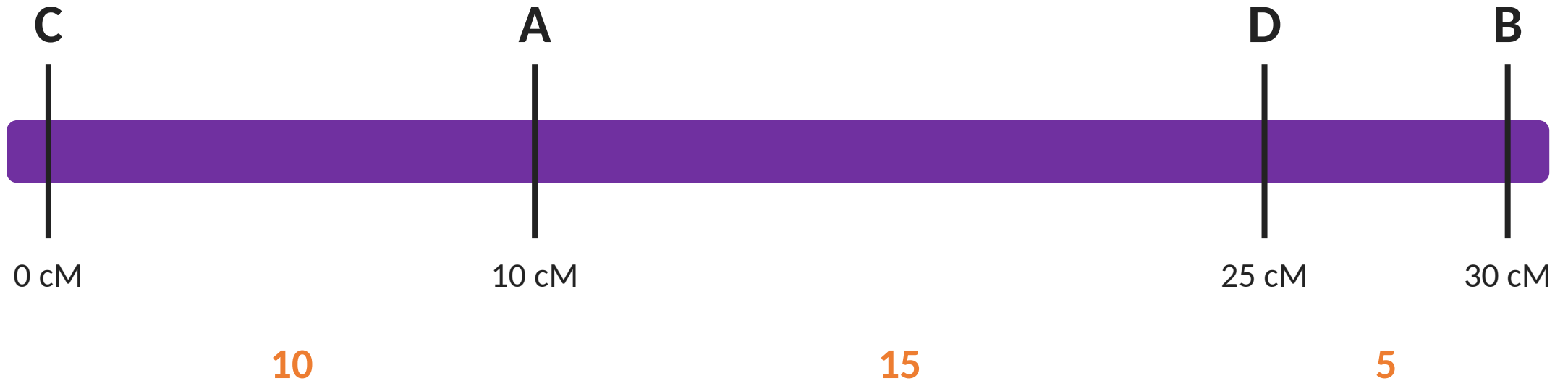
Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-maporder](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-maporder)

[host screen](#)

A linkage map

Order C – A – D – B. Distances add up: $10 + 15 + 5 = 30$



Repeat with hundreds of markers → a genetic map of every chromosome

Why the F2 is harder to read

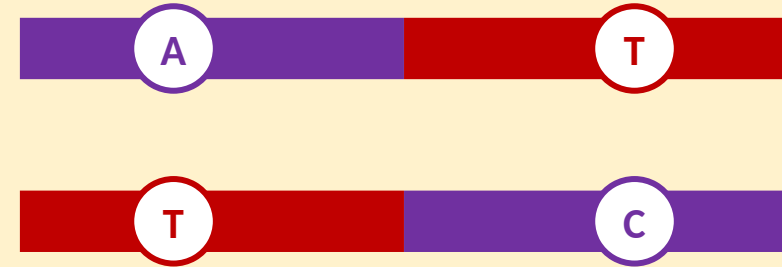
F2 offspring with SNPs AT CT : which story is true?

Story 1: two parental chromosomes



0 recombinants

Story 2: two recombinant chromosomes



2 recombinants

Same SNP genotype, different stories. That's why mapping uses crosses where one parent's chromosome is known (testcross, backcross) or inbred lines.

Why recombination matters

Finding genes

A marker that rarely recombines away from a trait sits near the gene.
This is how QTL mapping and GWAS work (coming later this quarter).

Breeding

A good allele drags its neighbors along ("linkage drag").
Recombination cuts off the bad hitchhikers.

Diversity

Every generation shuffles new haplotypes.
Why siblings differ, and why selection has new combinations to work with.

Play: the recombination game

Interactive linkage demo

Make crosses, collect offspring, count the recombinants, and build your own linkage map.



Scan with your phone

[Play on your phone or laptop: greymonroe.github.io/recombination-game/](https://greymonroe.github.io/recombination-game/)

Live poll

Text wall

Short Rht wheat always has many tillers. One gene doing two things (pleiotropy), or two tightly linked genes? How could you tell?



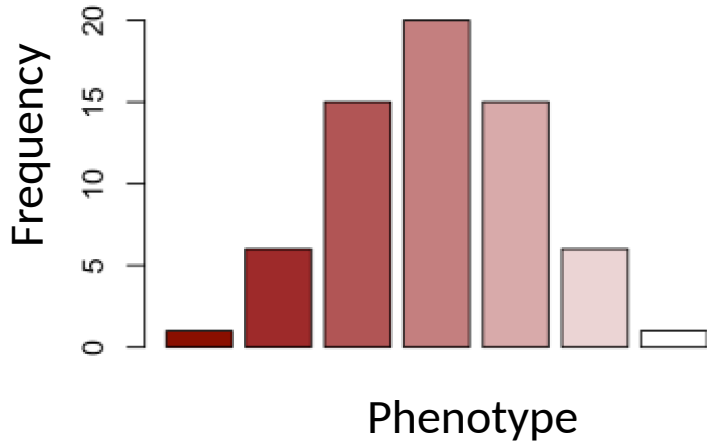
Scan with your phone

[Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-pleiolink](https://greymonroe.github.io/live-poll-test/submit.html?poll=pls152-tg2-pleiolink)

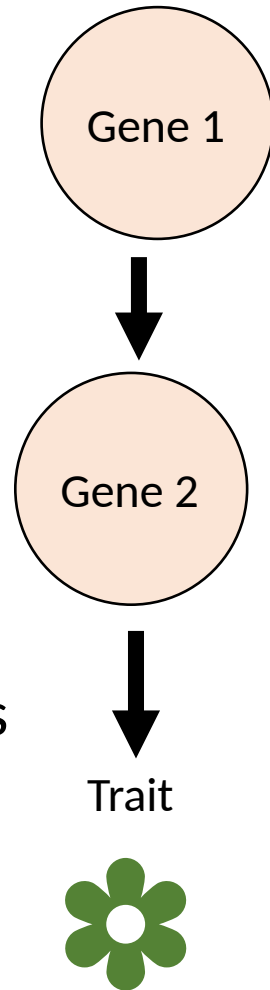
[host screen](#)

Most important take home messages

Incomplete dominance in many loci leads to **additive genetic variation** for quantitative traits.



Interactions between genes leads to **epistasis**

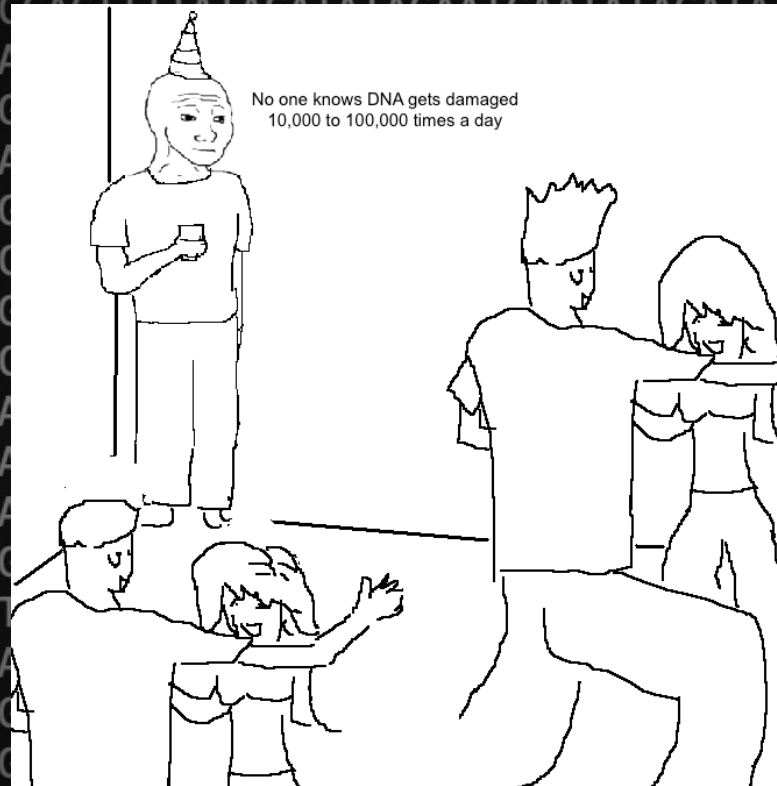


Physical linkage between loci on chromosomes leads to non-independent inheritance. **Close loci are inherited together (recombine less).**

A —

C —

Next week



The Biology of Alleles